Clue: A narrative writing puts an event in time order first, every boy ordered a mean then started waiting (the event sequenced).

Answer: A

110. Which one of the following sentences is correctly punctuated?
A. Send me Ketema, the headmaster; Jira, the unit leader; and Desta, the guard.
B. Send me ketema; the headmaster, Jira; the unit leader, and Desta; the guard.
C. Send me: Ketema, the headmaster; jira, the unit leader; and Desta, the guard.
D. Send me Ketema – the headmaster, Jira – the unit leader, and Desta the guard.

Clue: Use a comma after names and a semicolon after the job they are entitled.

Answer: A

111. We interviewed 10 managers of different branches in the city to see if what they think about the problem conforms with the information we secured from the customers. Which section of a study have the above lines been taken from?
A. Objective
B. Methodology
C. Introduction
D. Data Analysis

Clue: the writer uses one of data gathering tools – which is **interview**. (It is induced in research methodology).

Answer: B

112. We met at the back of the building and started looking for a quiet place where we could sit down and enjoy the presence of one another. The area was a bit dark and we were sure no one would come there in the mid of the night. We found a corner to sit and … these lines are taken from a piece of writing that is most likely:
A. Descriptive
B. Argumentative
C. Narrative
D. Expository

Answer: B

113. New situations always made me nervous. Which sentence is like to come next?
A. After a couple of minutes the teacher cam over.
B. My first swimming lesson was no exception.
C. I stood timidly by the side of the changing room.
D. I changed into my bathing suit.

Answer: B

114. Which one of the following sentences has correct capitalizations?
A. The united states has a Democratic form of government just as Great Britain does.
B. The United States has a democratic form of government just as Great Britain does.
C. The United States has a democratic form of government just as Great Britain does.
D. The United States has a democratic form of government just as Great Britain does.

Clue: Don't capitalize (government and democratic).

Answer: C

115. Which sentence is appropriately punctuated to avoid any ambiguity in meaning?
A. The doctor's wife said her friend is not well.
B. The doctor's wife said, her friend is not well.
C. The doctor's wife, said her friend, is not well.
D. The doctor's wife said, her friend is not well.

Clue:
- The reported sentence doesn't use an inside punctuations (comma, semicolon) but a full stop at the end.
- The apostrophe (') in "The doctor's wife" indicates possession. i.e., The wife belongs to the doctors. It should be **apostrophe + s** ('s).

Answer: A

116. Which one of the following sentences is most likely taken from a descriptive writing?
A. Unlike what they thought, we managed to get there early morning.
B. The whole area was wet and smoggy making driving extremely difficult.
C. In fact, it is good to start early morning if you want to drive long distance.
D. At 9 o'clock, some of the people started appearing from different directions.

Clue: A descriptive essay uses a vivid expressions so that the reader can visualize the image of an object or a person within their mind easily.
(Vivid word = wet and smoggy …)
↓ ↓
Word of touch word of seeing

Answer: B

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117. *She made me remember Mona Lisa. Her smiles and beauty were indescribable and as she came straight looking into my eyes, all I could do was stand dumbfounded. She got closer and closer and her eyes shone unbearably brighter...* This is taken from a piece of writing that is most likely:

A. Expository
B. Argumentative
C. Narrative
D. Descriptive

Answer: D (See Q 116)

118. *In conclusion, students from the rural area seem to have spent a significant portion or their time helping their parents whereas those from the urban centers hardly did anything of that kind.* What is the most likely research topic from which the preceding lines were taken?

A. An investigation into How Rural and Urban Students help their parents.
B. Residential area as a factor affecting the assistance students give to parents.
C. A survey of responsibilities of students from rural and urban areas in helping their parents.
D. Students from the rural and urban centers: how much time do they spend helping parents?

Answer: D

119. *When I looked up from the foot of the hill, I saw someone who was making almost to the tip and I knew it was Tesfaye. In no later than five minutes he stood on the top and started shouting down to those of us crawling up. ...* This is taken from a piece of writing that is most likely:

A. Narrative
B. Argumentative
C. Descriptive
D. Expository

Answer: A

120. "I have a proven track record of responsibility, integrity and commitment to company objectives." This is most likely taken from

A. A letter of application
B. An informal letter
C. A business letter
D. A letter of enquiry

Answer: A

1. Lung pollution is causes why air obvious cancer.

A. Why long cancer causes air pollution in obvious.
B. Lung cancer causes why air pollution is obvious.
C. Why air pollution causes lung cancer is obvious.
D. Why causes lung cancer air pollution is obvious.

Clues: The causer is air pollution- it causes lung cancer.

Answer: C

2. It to wanted you today me do.

A. Today you wanted it me to do.
B. It wanted me today you to do.
C. You wanted me to do it today.
D. Me wanted you to do it today.

Clues: **Subject – verb – object** is the right order of a sentence.
I (you (sub)- wanted (verb) me (object)

Answer: C

3. In got has house all money new the her she.

A. She has got all her money in the new house.
B. She has got her money all in the new house.
C. She has all her new money got in the house.
D. All her money she has got in the new house.

Answer: A

4. Planet could on our how that have beyond implications understanding life go far started earth.

A. How life started on our Earth could have understanding implications beyond planet that go far.
B. Understanding how life started on Earth could have implications that go far beyond our planet.
C. Earth could have understanding implications that go far beyond how our planet started on life.
D. Our planet could have implications how life go far started beyond that understanding on Earth.

Answer: A

5. That anything me don't to like again say.

A. Don't again say like that anything to me.
B. Don't to me say anything like that again.
C. Don't say anything to me like that again.
D. Don't say anything like that to me again.

Answer: C

Section Two: Paragraph Coherence (6 – 9)

6. a. It is about cultivating one's mind.
b. I believe in the power of education.
c. It also doesn't matter what color your skin is.
d. Learning can be the same for anyone – rich or poor.

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A. d e c a b C. b a e d c
B. d d a c a D. b d e c a

Clues: Topic sentence: **d/** supportive sentence: **e c a/** closing sentence: **b**

Answer: A

7. a. Such people do it more for pleasure than necessity.
   b. Weekend is the best time for those who buy essential things for life.
   c. Very few people can avoid shopping at least once a week.
   d. For some people, however, shopping has become a daily routine.
   e. Shopping is a necessary part of life.

A. e c d a b C. e d b c a
B. c d b a e D. b d c e a

Clues: Topic sentence: **e/** specific details: **c d a/** closing sentence: **b**

Answer: A

8. a. In 1991, the country launched an early – warning system – the first in the world and one that has been emulated by countries like Japan.
   b. Mexico's recent quake occurred 32 years after the city's 1985 shock, which killed about 10,000 people and destroyed 3,000 buildings.
   c. Since then, Mexico has worked to shore up its earthquake preparedness.
   d. Some had a minute to duck and cover, to grab go bags, to locate their loves ones.
   e. Thanks to sirens and smartphones, thousands of individuals were alerted to the impending quake.
   f. Those seconds can mean the difference between life and death.

A. a e b c f d
B. f a e b d c
C. c a b d f e
D. b c a e d f

Clues: Topic sentence: **b/** supporting sentences: **c a e d/** summing up sentence: **f**

Answer: D

9. a. I knew the journey could be tough but not as bitter.
   b. Something more unbearable! My visa application was denied.

   c. I migrated to Egypt in December 2005 dreaming to travel to Australia.
   d. In fact, you have to experience it yourself to see how severe things could be.
   e. The weather was indescribably harsh and the police were brutally unfriendly.

A. c a e d b C. c e a b d
B. c e d b a D. c a b e d

Clues: Topic sentence: **c/** specific details: **e a b/** closing sentence: **d**

Answer: C

PASSAGE I

THE SHRINKING EARTH: ECOLOGICAL THREATS

1. In 1977, the Food and Agriculture Organization (FAO) and UNESCO jointly published and alarming map of the spread of deserts across the world. Shaded in orange, pink and red were all the areas in danger of desertification. The colored patches covered a major part of the developing world outside the rainy equatorial belt. In Latin America they covered north – east Brazil, central and northern Mexico and stretched right down the Andes as far as Chile. The ; ate into the Horn of Africa and much of the south-west of the continent. And without interruption they reached half way round the globe in a broad swathe from the Atlantic coast of the Sahara, right across North Africa and the Sahel, through the Middle East and Persia to Pakistan and north-west India.

2. There could be no more graphic expression of the ecological dangers that threaten so much food production, so many livelihoods in the developing countries. In world is losing precious agriculture land at twice the rate that new land is being broken for farming. An area greater than Great Britain is disappearing every year. Soil is being exhausted, eroded, and blown away at the rate of two and a half billion tons per year. By the end of the century, the world may have to support one and a half times its present population on only three-quarters of its present cultivate area.

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3. In his report on the state of the environment in 1977, United Nations Environment Programme Director, Mostafa Tolba, warned that, if present trends continue, there would be only 0.15 hectars of farm land per person by the year 2000, half the 1975 level. Productivity would have to double merely to allow people to get the same amount of food as today. These are the figures on which he based that cultivation: in 1975 there were 1240 million hectares under cultivation. Over the next 25 years, perhaps another 300 million new under cultivation. Over the next 25 years, perhaps another 300 million new hectares may be opened up but over the same period 600 million hectares – half the entire 1975 cultivated area – may be lost. Of these, half will probably disappear under the **ink-blot spread** of cities, which are expanding horizontally twice as fast as the populations are growing, and over some of the best agricultural land. The other 300 million will be the toll of soil degradation.

4. At least half of the total erosion will be in the world's 45 million square kilometers of potentially productive but ecologically precarious dry lands, which stretch through a hundred nations. About 700 million people live in this zone, almost all them in developing countries, and 18 million live in areas that are currently undergoing rapid desertification.

5. Everywhere the deserts are advancing. In Sudan, the southern edge of the Sahara moved south by 100 kms between 1958 and 1975. The deserts do not march forward on a solid front, like an army. Patches appear, like those at Aorema in Burkina Faso, around centers of population or watering holes, then spread, link up with others, and finally merge into the desert itself.

6. The chief agent of what has been called **leprosy of the soil** is man, the impact of his activities on highly sensitive and delicately balanced ecosystems. The **prime** factor in the process is population increase. The number of people in the Sahel, for example, is doubling every 25 to 30 years. We

can see everywhere the consequences of this among settled farmers: the cutting down of fallow periods, progressive decline in the vegetation cover, increasing erosion. As population goes on growing, cultivation is pushed into areas that are entirely unsuitable for agriculture, and there the process progresses even more rapidly.

Taken from: English Four, ETC, Department of English

10. How many African countries have specifically been mentioned as those in danger of desertification?
A. 2 B. 3 C. 4 D. 5
Answer: A

11. What is the main idea conveyed in paragraph 1
A. Africa has been more seriously affected by desert than Latin America.
B. The areas outside the rainy equatorial belt have not been affected by deserts.
C. Deserts affected developing countries more than they did the developed ones.
D. FAO and UNESCO have shown a serious concern about desertification problems.
Answer: D

12. According to paragraph 2, how much of our soil is getting out of use every six months?
A. 1.25 billion tons C. 1.50 billion tons
B. 2.50 billion tons D. 1.75 billion tons
Answer: B

13. Which paragraph could be cited as a good example to show that the continent of Africa has been seriously affected by desertification?
A. Paragraph 1 C. Paragraph 5
B. Paragraph 4 D. Paragraph 6
Answer: C

14. In relative terms, which one of the following factors contributes more to desertification?
A. Increase in the amount of erosion.
B. Decrease in land left uncultivated.
C. Decline in vegetation covering the areas.
D. Cultivation of land unsuitable for farming.
Answer: A

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3. In his report on the state of the environment in 1977, United Nations Environment Programme Director, Mostafa Tolba, warned that, if present trends continue, there would be only 0.15 hectars of farm land per person by the year 2000, half the 1975 level. Productivity would have to double merely to allow people to get the same amount of food as today. These are the figures on which he based that cultivation: in 1975 there were 1240 million hectares under cultivation. Over the next 25 years, perhaps another 300 million new under cultivation. Over the next 25 years, perhaps another 300 million hectares may be opened up but over the same period 600 million hectares – half the entire 1975 cultivated area – may be lost. Of these, half will probably disappear under the **ink-blot spread** of cities, which are expanding horizontally twice as fast as the populations are growing, and over some of the best agricultural land. The other 300 million will be the toll of soil degradation.

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D. FAO and UNESCO have shown a serious concern about desertification problems.

Answer: D

12. According to paragraph 2, how much of our soil is getting out of use every six months?
A. 1.25 billion tons C. 1.50 billion tons
B. 2.50 billion tons D. 1.75 billion tons

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A. Paragraph 1 C. Paragraph 5
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Answer: C

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A. Increase in the amount of erosion.
B. Decrease in land left uncultivated.
C. Decline in vegetation covering the areas.
D. Cultivation of land unsuitable for farming.

Answer: A

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15. Which one of the following sentences best summarizes paragraph 2?
   A. The world is losing its precious agricultural land at an alarming rate.
   B. Soil degradation is a serious problem threatening people in developing countries.
   C. The amount of cultivable that land is decreasing while population remains constant.
   D. The amount of cultivable land and population growth are inversely proportional.

Answer: B

16. What does the phrase, the **ink-blot spread** of cities (paragraph 3) mean?
   A. Spread of cities along the main roads.
   B. Establishment of mega cities on farmlands.
   C. Establishment and spread of cities into the farmland.
   D. Establishment of a city to be followed by another in short distance.

Answer: D

Passage II

1. UNITED NATIONS – Nearly 60 million people have been driven from their homes by war and persecution, an unprecedented global exodus that has burdened **fragile countries** with waves of new comers and littered deserts and seas with the bodies of those who died trying to reach safety. The new figures, released recently by the United Nations refugee agency, paint a staggering picture of a world where new conflicts are erupting and old ones are refusing to subside, driving up the total number of displaced people to a record 59.5 million by the end of 2014, the most recent year tallied.

2. Half of the displaced are children. Nearly 14 million people were newly displaced in 2014, according to the annual report by the office of the United Nations High Commissioner for Refugees. In other words, tens of thousands of people were forced to leave their homes every day and "seek protection elsewhere" last year, the report found. Tens of millions of others fled in previous years and remain stuck, sometimes for decades, unable to go home or find a permanent new one, according to the refugee agency. They include the more than 2.5 million displaced in the Darfur region of Sudan, and the 1.5 million Afghans still living in Paksitan.

3. When refugees flee their own countries, most of them wind up in the world's less-developed nations, with Turkey, Iran and Pakistan hosting the largest numbers. One in four refugees now find shelter in the world's poorest countries, with Ethiopia and Kenya taking many more refugees than, say, Britain and France. As the report states, "the global distribution of refugees remains heavily skewed away from wealthier nations and towards the less wealthy." Even so, there has been a sharp backlash in European Union has shelved its plans to get approval from the United Nations Security Council to target human smugglers who operate in lawless Libya and to destroy the ships they use to bring migrants across the sea. Instead, the European Union is scheduled to meet on Monday to discuss whether it will start military operations in the international waters of the Mediterranean Sea, for which it does not need the Council's blessings.

4. European Union leaders are still squabbling with one another over how to split up **at least 40,000 asylum seekers** across their 28 member states. And they have stepped from a sharp-and-rescue operations after intense public pressure stemming from a sharp increase in the deaths of migrants crossing the Mediterranean this year. Australia has felt no such compunction. Its prime minister, Tony Abbott, has pledged to turn around migrant boats before they enter Australian territorial waters, including those with minority Rohingyas fleeing persecution in Myanmar. His administration **faces scrutiny over allegations** that it paid smugglers to turn a boat back to Indonesia after it was intercepted on the high seas.

5. "For an age of unprecedented mass displacement, we need an unprecedented humanitarian response and a renewed global commitment to tolerance and protection for people fleeing conflict and persecution," Antonio Guterres, the high commissioner for refuges, said in a statement accompanying the annual report.

6. Amnesty International, in a **report issued this week**, accused governments and smugglers alike of pursuing "selfish political interests instead of showing basic human compassion." The United States offers permanent resettlement to roughly 70,000 refugees each year, though its plans to admits 2,000 Syrian refugees this

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year have drawn scrutiny from Republican lawmakers who worry that some among them might be terrorists. The war in Syria is the largest source of displacement. By the end of 2014, 7.6 million Syrians were displaced within the country itself, and nearly another 3.9 million were refugees living outside the war-torn nation.

Source: United Nations English Language Programme

17. Which countries in paragraph 3 are among those that should take the blame with regard to creating refugees?
A. Britain and France
B. Ethiopia and Kenya
C. Syria and Eritrea
D. Turkey, Iran and Pakistan
Answer: C

18. What made the European Union intensify its search – and – rescue operations?
A. The sharp increase in death of migrants.
B. The increasing number of asylum seekers.
C. Their intention to send back new arrivals.
D. The serious demand put to them by the public.
Answer: A

19. What does the phrase 'fragile countries' (paragraph 1) refer to?
A. Countries located around deserts and seas.
B. Countries that had to host emigrants from other places.
C. Countries from where people are persecuted for some reason.
D. Countries from where people leave their homes because of war.
Answer: D

20. Which one of the following sentences reflects Antoni Guterres' view towards refugees?
A. People are being persecuted at an unprecedented rate.
B. People who flee conflict and persecution need to be more tolerant.
C. The world should be more tolerant to those displaced from their homes.
D. We need to be aware that there is an unprecedented humanitarian response today.
Answer: C

21. Why is the United States reluctant to admit Syrian refugees?
A. Because it runs its own political interests.
B. Because there are terrorists mixed with the refugees.
C. Because Republican lawmakers do not trust the refugees.

D. Because it is possible for the refugees to settle within Syria.
Answer: B

22. According to paragraph 2, which one of the following sentences is **NOT** true?
A. The 1.5 million Afghans living in Pakistan do not seem to want to go back home.
B. Compared to the adolescent or the elderly emigrants, children are larger in number.
C. Newly dislocated people in 2014 make up nearly a quarter of the total number of displaced people.
D. Relatively a significant number of those who flee their countries fail to find permanent homes.
Answer: A

23. What is the paradox that one can see in paragraph 3?
A. Wealthy countries take more refugees than the poor ones.
B. Poor countries take more refugees than the wealthy ones.
C. Wealthy countries were supposed to take more refugees than the poor ones.
D. People who flee their own countries eventually land in less-developed nations.
Answer: B

DIRECTIONS: Questions 24 – 36 are incomplete sentences. There are four alternative words/phrases, A-D, given below each question. Choose the one that best completes each sentence and **blacken** the letter of your choice on the separate answer sheet provided.

24. An increase in the prevalence rate of HIV/AIDS incidence reveals that some people are still _____ to their sexual partners.
A. ungrateful
B. unkind
C. unfaithful
D. unthankful
Clues: If you cheat your sexual partner/ wife, you're said to be unfaithful (not faithful) to her.
Answer: C

25. As **docile** is to **tamed** so is **awesome** to _____.
A. distasteful B. fearful C. blunt D. grand
Clues: **docile** and **tamed** are synonym words- meaning manageable. Similarly, **awesome** would be synonym to **fearful** – frightening
Answer: B

26. They were two loving sisters. Nobody could tell why they fell _____.
A. through B. apart C. in D. out

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Clues: "... were two loving sisters" means "they loved each other so much." But now they do not. They **fell out**. **Fell out** means **quarreled**. **Answer: D**

27. Adamu was given enough time to do his assignment; however, as usual, he came _____ several excuses for failing to do his work.
A. up with B. across to C. through D. in
Clues: if someone **provides** reasons or excuses for failing to do something, they **come up with** it. **Answer: A**

28. We were not happy with the hotel room we stayed in. Contrary to their advertisement, it was noisy and the views were not nice. We have to _____ a claim to be at least partially refunded.
A. Put out C. Put up
B. Put through D. Put in
Clues: They want to **request** the hotel manager a claim for they are discomforted with their stay in that hotel. **To request** a claim means **to put** in a claim it's a phrasal verb. **Answer: D**

29. Underprivileged or _____ children deserve special consideration. We need to make decisions on how to assist them.
A. disabled C. disadvantaged
B. uneducated D. inaccessible
Clues: The conjunction **'or'** after the word 'underprivileged' tells us the new word needs to be filled in the blank space is similar/ synonym to it. So, **underprivileged** means **disadvantaged** society that have less money and fewer opportunities than most people. **Answer: C**

30. I know she is lying, but it is hard to _____ her story.
A. disclose C. dismantle
B. disprove D. discourage
Clues: to disprove: to show that something is wrong or false. (In spite of lying, it is hard to say that her story is wrong/false: it's hard to disprove...) **Answer: B**

31. I'm meeting Sara at three o'clock to discuss our class schedule. After that, I'm playing chess. Oh, _____, I may want something to eat when I get home.
A. by the way C. whereas
B. after all D. besides **Answer: B**

32. No translation; students need to learn to _____ what words mean from the way they are used.
A. reduce C. induce
B. introduce D. deduce
Clues: If you understand the meaning of words based on their context in a sentence, you're said that you **deduce** them. (You make deduction, you **infer** them) **Answer: D**

33. Physical labor can exhaust the body; _____ excessive reading can reduce mental power.
A. regrettably C. equally
B. incidentally D. sadly
Clues: The impact of physical labor on the body and the excessive reading on mental power is equally explained. (In the same ways/ at the same time/ similarly = equally) **Answer: C**

34. The ceremony did not take long; I _____ the statue soon after the coordinator's brief speech.
A. unveiled C. unwrapped
B. unsealed D. unrevealed
Clues: to **unveil** means to remove a cover or certain from a painting, statue, etc. so that it can be seen in public **Answer: A**

35. Children using the swimming pool must be _____ by an adult at all times.
A. adapted C. accustomed
B. accorded D. accompanied
Clues: This means children should go swimming pool with adults. (They must be **accompanied** by an adult) **Answer: D**

36. I usually get nervous at the beginning of interview sessions but as time goes, I become _____ and get back to normal.
A. stable C. calm
B. settled D. healthy
Clues: Being not excited/ nervous/ upset is being **'calm'** **Answer: C**

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B. Substitution (37 – 40)

DIRECTIONS: Question 37 – 40 each has an underlined word or phrase. There are four alternatives, A-D, given after each sentence. Choose one word or phrase that **best** keeps the meaning of the given sentence if substituted for the underlined word or phrase and **blacked** the letter to your choice on the separate answer sheet provided.

37. I always believe it is healthy for everyone to repress their emotions.
A. Hold back C. Throw away
B. Throw out D. Kick out
Clues: to **repress** means to force fully prevent an upheaval from developing further. (= to keep back) to restrain to **hold back.**
Answer: A

38. Actually, she takes after her mother, not her father.
A. Resembles C. Remembers
B. Cares for D. Follows
Clues: If you **take after** one of your parents (Mother/ father) you **resemble** them in your physical appearances. **Answer:** A

39. The dog went for me when I knocked at their door this morning.
A. Barked C. Growled
B. Attacked D. Grunted
Clues: Went for is the past form of **go for** which means attacked/ hurt. **Answer:** B

40. We heard the news and called off the meeting.
A. Postponed C. Cancelled
B. Held D. Chaired
Clues: to call off a meeting or a program means **to cancel** it/**to put off** a meeting or a program means to postpone it to another day. **Answer:** C

GRAMMAR (41 – 66)

DIRECTIONS: Questions 41 – 66 are incomplete sentences. There are four alternative words or phrases, A-D, given below each question. Choose the world or phrase that **best** completes the sentences and **blacken** the letter of your choice in the separate answer sheet provided.

41. I completely _____ with you; taxi drivers and waiters shouldn't expect tips.

A. disagree C. disagreed
B. have agreement D. have disagreed
Clues: Use the present simple to talk about the present time.
Answer: A

42. Everyone knew about the change in the exam schedule, but _____ of them told me about it.
A. some B. none C. few D. any
Clues: The structure after **"but"** should be the opposite idea that is none of them told me about it. **Answer: B**

43. I _____ any good films for the last six months,
A. had not seen C. have not seen
B. did not see D. i don't see
Clues: To express the action began in the past and continues up to present, present perfect is used. Because of **'any'**, the **negative stricture** should be used. **Answer: C**

44. _____ you score an **A** on this exam, you will not choose to study English in the university.
A. Because B. If not C. Except D. Unless
Clues: use '**unless**' to say 'If you don't do…' because **unless** has the meaning '**if … not'.** **Answer: D**

45. We all should learn to be more kind _____ each other.
A. to B. for C. with D. at
Clues: some adjective like good, nice, polite, rude, kind, important, etc are used with 'to'.
- You should always be **polite/ kind to** people.
- My family is **important to** me. **Answer: A**

46. Let's get going; I have _____ heard such nonsense.
A. ever B. forever C. never D. wherever
Clues: Adverbs of time such as ever, never, already, just, yet are usually used with present perfect in different ways.
- Have you ever done such miracle thing before?
- I have never visited such nice place before.
Answer: C

47. The money we collected is _____ near enough to cover our expenses. Much more is needed still.
A. anywhere C. somewhere
B. everywhere D. nowhere
Clues: use **any** and **no** with their compound words in different patterns. **Anything/ Anywhere/ Anyone… nothing/**

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no where/ no one... Negative word + **any**...
Positive word + no...

- I couldn't find my pen **anywhere** in this classroom. (could not + any where)
- I could find my pen **nowhere** in this classroom (could + **nowhere**)

Answer: D

48. How _____ a three-year-old child be expected to tie his shoelaces without any help?
A. may B. can C. must D. has
Clues: In this sentence you can understand inability (not to be able to do something)
- **How can** I lift up the 100 kg weight? (I can't lift up/ I am unable to lift up...?)

Answer: B

49. I don't choose teaching as my profession, _____ I know teachers have long vacations.
A. but B. therefore C. and D. so
Clues: The first sentence has the opposite meaning with the second one.
Answer: A

50. I don't understand why she wouldn't lend us one if she _____ two books.
A. has B. has had C. had D. had had
Clues: In unlikely conditional clause: If + past tense and would + v_1 are used.
- I would lend you some money if I received my salary.

Answer: C

51. We must work hard _____ to extricate our country from poverty.
A. in order C. such that
B. so that D. because
Clues: Use 'to/ in order to/ so as to + V_1 to express a purpose clause.
Use **so that/ in order that + clause** to express the same clause.
- I study hard **so as to do** well on the exam.
- I study hard **in order that** I can do well.

Answer: A

52. _____ we grow older, w care more about what other people think about us.
A. While B. As C. Because of D. In case of
Clues: When we say that two things always **go parallel**, we can use 'as.'

- As we grow older, we care more about what other people think about us
(= The more we grow older, the more we care ...)

Answer: B

53. I was hoping to find some funny stories on your shelf, but I did not see _____.
A. many B. a few C. any D. some
Clues: I didn't see **any**/ I did see **none**.

Answer: C

54. Which of the three deserves the school's prize _____?
A. most B. well C. best D. better
Clues: To compare more than two things/ people, use superlative form of adjective. (from the three deserves, the one is the best, the question implies.)
- Who is **the best/ least character** form the four?

Answer: C

55. My mother caught the knife carefully _____ not to cut herself.
A. so that B. to C. in order D. because
Clues: To express a purpose sentence in a negative way we use **not to/ in order not to/ so as not + V_1.**
- Please open the door quietly **in order not to/ so as not to disturb** the asleep baby.

Answer: C

56. The meeting had not begun _____ when we arrived.
A. already B. for C. since D. yet
Clues: we use 'yet' as an adverb time with present perfect to say "up to now" It often takes the end position of present perfect.
- By the time I left home, she hasn't weaken **yet.**

Answer: D

57. We have known each other _____ childhood.
A. when B. while C. since D. as
Clues: Use **since** + the starting period of an event in the past.
since + 2010 **since last year**
since + September since the school close
(= present perfect + since + past event)
- I have stayed with my uncle **since Christmas.**

Answer: C

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58. Guess how long you _____ at school by the end of this year
A. will be C. had been
B. have been D. will have been
Clues: Use future perfect tense "**will have been**" to talk about the action that will be over before another action in the future
Answer: D

59. I'm not being superstitious, but you _____ something earlier that brought you this bad luck.
A. should have to do C. must have done
B. might have been doing D. would have done
Clues: The person who is spoken to likely faced a bad luck and the person who is speaking is sure that there is something done that brought that person bad luck. Such idea is introduced **with past deduction = must have + V_3**
X: I am not feeling comfort after the last night's party.
Y: Oh, you **must have drunk** much.
Answer: C

60. There are many occasions when seat-belts save life, yet only _____ drivers often wear them.
A. many B. a lot of C. a few D. few
Clues: a few means some amount of number, but not many; **few** means hardly any amount of number. It's negative so, not all drivers tie a safety belt, only a few of them do
Answer: C

61. The jury had no choice, but to return a verdict of guilty _____ with all the evidence.
A. when presented C. while they were presenting
B. as they presented D. after they presented
Clues: As they presented means **as soon as**/ immediately after they are presented. Thus, As they presented is better in this case.
Answer: B

62. Nobody would like to go with a _____.
A. team loose C. team lost
B. lost them D. losing team
Clues: We can use past and present participle to describe different things.
- **losing** generation = the generation which is in the sequence of losses
- **lost** generation = the generation that came of age

Answer: D

63. The decision of the committee was fair; they offered the scholarship to all _____ students.
A. deserved B. deserving C. deserve D. Deserver
Clues: deserved (የሚገባው/ የተገባው)
Three deserved students were awarded.
The deserved students were awarded **Answer: A**

64. What an awful thing! I thought the weather would be _____ in April, but it actually got _____
A. worse/better C. better/worse
B. good/better D. good/worse
Clues: Two opposite adjectives are needed in these blank space **better** and **worse**.
- Yesterday I felt **better**, but today I feel worse. **Answer: C**

65. We _____ at the scene when the alarm went off.
A. have just arrived C. had just arrived
B. were just arriving D. arrived
Clues: Past tense and past continuous tense often go together. The past continuous tense is the action going on, and the past simple is the action happened then.
- **I was reading** a story when I heard a shocking news.

Answer: B

66. By the end of August, we _____ our placement in universities.
A. will have known C. will know
B. will have been known D. will be knowing
Clues: Key phrases such as **by June, tomorrow this time, at the end of the term, etc.** are used to express **future perfect tense**.
- **I will have finished** my work when you are back from school.

Answer: A

COMMUNICATIVE ACTIVITIES (67 – 98)

DIRECTIONS: Questions **67 – 98** are presented in a form of dialogue. The part said by one of the speakers is given, and a blank space is let for the other. For each of the blank spaces, four alternative answers, **A-D**, are given. Choose the one that is most appropriate to complete the dialogue and **blacken** the letter of your choice on the separate answer sheet provided.

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67. **Temesgen**: There is an exhibition at the stadium.
Abera: What's it on?
Temesgen: Charity
Abera: It may be interesting. Are you planning to go?
Temsgen: _____
A. I have a tight schedule, but I may try.
B. How do you know?
C. Many people are going.
D. It will surely be interesting.
Answer: A

68. **Samarawit**: Do you know how to make pizza?
Konjet: _____
A. It is easy
B. I don't like pizza
C. No, I don't
D. Yes, I can
Clues: The reply can either be "Yes, I do or No, I don't."
Answer: C

69. **Husband**: There is someone at the door.
Wife: Wait, you had better put your shoes _____
A. On before you open it.
B. On after you open it
C. Off before you open it
D. Off after you open it.
Clues: put a cloth **on** means **dress** it.
put a cloth/ shoes **off** means **remove it.** } Phrasal verbs
Answer: A

70. **Daniel**: I got the job I told you about.
You: _____
A. I don't care
B. Congratulations!
C. That's good news
D. Jobs are not easy to get these days.
Clues: If your friend tells you something interesting about her/him, for example, getting a good job, winning chances, etc. you'll express your happiness to him/her saying **'congratulations!'**
Answer: B

71. **Tamrat**: Do you know who is standing over there?
Lema: _____
A. No, I don't
B. Who is he talking too?
C. He is very tall, isn't he?
D. I met him yesterday
Clues: If you know, say "Yes I do," if not, say "No, I don't"
Answer: A

72. **Chaltu**: What begins with **T**, ends with **T** and has **T** in it?
Genet: _____
A. Tea party
B. Tinted pot
C. Teapot
D. Tantamount
Clues: Teapot: is also called **teakettle** and it is for serving tea.
The logic is this: * Tea: begins with T
* POT = ends with T
* contains Tea = T
Answer: C

73. **Aunt**: _____
Lojo: Sure
A. Pass me the bread, will you?
B. Give me the bread.
C. You will give me the bread.
D. I want the bread.
Clues: You say **'sure'**, when you are asked for something and you are **willing/ free** to do that.
Answer: A

74. **Jemal**; What do you make of this painting, Jemila?
Kedija: _____
A. Nothing
B. Everything
C. Something
D. Anything
Clues: The other way to say this is "What is this painting made of Jemila?"
Answer: D

75. **Getahun**: What was your holiday like?
Fetiha: _____
A. We liked it very much.
B. Kedija came to our house.
C. It was nice but the weather was not good.
D. Rarely so!
Clues: What was your holiday like = how was … like?
Answer: C

76. **Azeb**: Have you heard about the MDGs?
Haben: Yes. I have
Azeb: Do you think they can be achieved?
Haben:_____
A. There is abundant research on MDGs.
B. Developed countries are excluded.
C. I'm not sure they are based on practical experiences.
D. There are people who accept or reject the MDGs.
Answer: C

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77. **Fetle:** Can we practice solving riddles?
Kassech: Ok. You start
Fetle: I come once in a year and twice in a week. Who am i?
Kassech: But his is illogical. In a week once, in a year twice! Not all riddles follow logic for their solution. Some encourage you to look for other means.
Kassech: I cannot solve your riddle.
Fetle: _____
Clues: There is one **'e'** in the word year and two **'e's'** in the word week.
Answer: C

78. **Fasil:** Are you scared of girls?
Ibrahim: Yes, Fasil. _____
A. I can't think of anything to say.
B. There are many things to say.
C. I know what to do.
D. I have four sisters.
Clues: "Yes, Fasil" should be followed by a **positive idea**.
Answer: B

79. **Mother:** May I ask you who can take part in your 'protect yourself' program?
Daughter: Anyone who wants, but _____.
A. Rearely it is young women
B. Interested individuals
C. You aren't a girl
D. Mostly it is young women
Answer: D

80. **Friend:** Your cousin has just told me he's getting married next week. Are you very pleased?
You: _____, but he could have told me earlier.
A. No, actually
B. I'm not
C. I'm absolutely delighted.
D. Where is the wedding to be held.
Answer: B

81. **Father;** _____
Son: But it's good for health
A. Physical exercise is good.
B. Do you like physical exercise?
C. Is physical exercise good?
D. Physical exercise is tiring.

Clue: The two clauses should be contrast each other.
Physical exercise is tiring, but it's good for health
Answer: D

82. **Friend:** _____.
You: You should loosen your belt.
A. I have got stomachache.
B. I ate too much
C. I am terribly hungry
D. I am starved to death
Clue: If you eat too much food, your stomach will become bigger.
Answer: B

83. **Receptionist:** I get mad when someone rings me up late at night.
Guest: _____
A. Yes, that can be annoying, but you should try to let it upset you.
B. Yes, that can be annoying, but you should try not to let it upset you.
C. No, that can be annoying, but you should try to let it upset you.
D. No, that cannot be annoying but should try to let it upset you.
Answer: B

84. **Amir:** They took my money, my credit cards, my passport, everything.
Dano: _____
A. Oh dear! What are you going to do now?
B. You should see a doctor.
C. Why are you here now?
D. Why not? Go to the police.
Answer: D

85. **You:** How can I get to the nearest pharmacy, please?
Ahmed: _____
A. Why do you ask?
B. How long have you had this headache?
C. Go straight and turn right.
D. Do you want to see a doctor?
Clues: This question is about finding the way/ direction.
Answer: C

86. **Nardos:** I heard were the only person to get the job. Well done!
Girma: _____
A. I knew, I did well. C. Why expected I would not?

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B. Sure, I did. D. Thank you.
Clues: The replies to "well done! Congratulations!" can be thanks a lot! **Thank you very much…"** **Answer: D**

87. **Sara:** I have such a bad headache that I can't even open my eyes.
Hana: _____
A. Where is the nearest clinic?
B. Pherhaps you need some break.
C. Do you often have it?
D. I have headaches. **Answer: B**

88. **Teacher:** You did not attend class yesterday, did you?
Student: _____
A. No, you didn't C. No, I did
B. No, you did D. No, I didn't
Clues: The answer after positive tag is negative sentence.
X: Your mother is a nurse, isn't she?
Y: Yes she is.
X: Your brother didn't pass the exam, did he?
Y: No, he didn't. **Answer: A**

89. **Student 1:** Virus attacks are common and the spread of anti-virus might tip the balance towards treating them seriously.
Student2: I cannot follow you. Can you please paraphrase it?
Student1: _____
A. How does the virus steal your documents?
B. In other words, attention would be given to arresting the effects of the virus.
C. There is always an antivirus available to protect your computer.
D. Wanna Cry is a virus that steals documents in your computer. **Answer: B**

90. **Deboch:** National exams should not continue.
Teklu: Probably, you may be right.
Deboch: Since we cannot do away with exams, regions may need to set their own.
Teklu: This may be a good idea, but there is the issue of standard.
Deboch: The Federal Ministry may set the basic standards.
Teklu: _____
A. Well, I may agree with you.
B. I hate exams.

C. Exams set standards.
D. There are ways and methods.
Clues: Teklu partcally agrees with Deboch. **Answer: A**

91. **Guest:** What a room service! There is no towel in the bathroom. Could you please bring me one?
Host: Yes, madam. I will bring one soon.
Guest: _____ The shower does not function. Can you please fix it?
Host: We're sorry madam. The plumber is on leave. If you don't mind, could you use the common shower next door?
A. Alas!
B. Terrible! I don't understand.
C. Oh my God! What a curse.
D. Where's the plumber? **Answer: D**

92. **Gennet:** Our class is planning for a picnic next week. Will you join us?
Chaltu: well, I don't know. I have to get permission from my parents.
Gennet: But try hard. Don't miss it.
Chaltu: _____
A. I don't miss it
B. My parents like it very much.
C. I miss it.
D. I may come.
Clues: It's doubtful that Chaltu may be given permission from her parents. **Answer: D**

93. **Dandir:** Is Addis Ababa getting more and more dangerous? People say there are gangs everywhere.
Gemechu: No, no._____
A. This kind of behavior simply won't be tolerated
B. It's simply a misunderstanding
C. It's simply an exaggeration
D. That's simply the best the city can afford **Answer: B**

94. **Son:** Oh no! _____
Father: I will bring you a mob.
A. Has the bathroom flooded?
B. The bathroom flooded.
C. What has happened?

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D. The window is open.
Clues: a mob (mob cap) is a light cotton cap covering all the body. That is why the father says "I'll bring you a mob."
Answer: D

95. **Daughter:** _____
Mother: You should go on a diet.
A. I am putting on a lot of weight.
B. Tell me the best restaurant here.
C. I have no appetite.
D. I am losing weight.
Clues: someone who is losing weight is advised **to go on** a diet.
Answer: D

96. **Agaredech:** Global warming and extreme pollution are interconnected.
Behailu: But some argue that the earth is getting aged.
Agaredech: While this may have some truth, pollution may have hastened the process.
Behailu: Certainly, but there are global efforts to curb pollution.
Agaredech: _____
A. True. But the link between the two should be unequivocally established.
B. There is no pollution in backward countries.
C. Global warming is inevitable.
D. Environmentalists tried to influence big international organizations.
Answer: D

97. **Guest:** The kitchen handle has come off.
You: Don't worry. _____
A. It has never been like that
B. It often does
C. You will buy a replacement
D. It's none of you business
Clues: Come off (phrasal verb): to become detached. **Answer: C**

98. **Mother:** Don't you ever run out into the road like that. You could have been knocked over.
Daughter: _____
A. How come that this ever happens to me?
B. What? You could have told me that earlier.
C. Oh no! Thank you, Mom.
D. Have I ever been knocked over?
Answer: C

WRITING (99 – 120)

DIRECTIONS: Questions 99-120 are related to different types of writing. Read each of them very carefully and choose the best answer from the four alternative answers, A-D, given. Then blacken the letter of your choice on the separate answer sheet provided

99. Which one of the following is taken from a formal letter of application?
A. I am writing in response to the position of IT assistant advertised in the Monitor.
B. Thank you for your letter dated 5th April which I received this morning.
C. I would be grateful if you could confirm your reservation in writing.
D. Sadly, I am not available on the date you suggested. May I suggest the 8th of April?
Clues: The kind of formal letter used to find a job is called application letter. **Answer: A**

100. Which of the following contains **errors in capitalization, mechanics and punctuation**?
A. The fire, although it had been burning for several hours, was still blazing fiercely.
B. You should, indeed you must, report the matter to the police.
C. "When the judge said, 'Not guilty,' I could have hugged him."
D. Peter, my elder brother received medals for his contributions Astronomy, Chemistry and Mathematics.
Clues: It is only the phrase "Not guilty" that should be enclosed in quotation mark. **Answer: C**

101. Which one of the following is taken from an **informal letter?**
A. I would like to apply for the position of assistant secretary.
B. I enclose my curriculum vitae for your attention.
C. Wish me good luck for my math exam.
D. I am writing to complain about the following damages.
Clues: An informal letter is a friendly/ personal letter. It is not as tight as a formal/ business letter. **Answer: C**

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102. "... He couldn't understand where he had gone wrong. When he had set out that morning, the weather had been fine..." The preceding text is possibly taken from a(n)_____writing.
A. descriptive
B. expository
C. narrative
D. argumentative
Clues: The type of writing telling us a story or narrates what/ when/ where, etc happened is known as Narrative essay.
Answer: C

103. Which one of the following is **correctly punctuated**?
A. I have never been to England; in fact; I have never been outside this country.
B. I have never been to England; in fact I have never been outside this country.
C. I have never been to England, in fact, I have never been outside this country.
D. I never been to England; in fact, I have never been outside this country.
Clues: When we use the phrase "in fact" we should use a longer pause mark (semicolon) before and a short pause mark (comma) after it. (...; in fact,...)
Answer: D

104. Which one of the following sentences has correct capitalization?
A. The language I wanted to learn was Arabic, not French.
B. The language I wanted to learn was arabic, not French.
C. The language I wanted to learn was arabic, not french.
D. The language I wanted to learn was Arabic, not french.
Clues: The name of the languages should be capitalized
Answer: A

105. *"In the meantime, it is necessary to find other ways of disciplining drug offenders."*
The above line was taken from the recommendation section of a survey on drug conviction. Which of the following is true about the recommendation?
A. This is the major point recommended in the study.
B. It is one of the two recommendations in the study.
C. This is recommended to take place immediately.
D. It is a recommendation forwarded until better actions can be taken.
Answer: D

106. *"Of late, drugs have changed American penal practice."*
The above line was taken from the introduction section of a study carried out on drug offenders in America. Which of the following sentences should come next to the quotes sentence?
A. Nations with different penal practices have experienced different crime rates.
B. In 1987, for example, only about 8 percent of state-prison inmates were serving time on drug convictions.
C. People were rightly fearful of what cocaine was doing to their children when they demanded action.
D. It needs to be pointed out that there are no easy prison solutions to drug convictions.
Answer: B

107. *"You may not need to write your name."* Where do you find this statement in a survey study?
A. It can be one of the items in the questionnaire.
B. It is a reminder written in questionnaire coveting letter.
C. It is taken from respondents' personal information.
D. It is taken from a 'Thank you' letter written to respondents.
Answer: B

108. *"Dressed purposely to confuse her friends in the first week of school, Samrawit wears clothes she would normally never wear. With her new haircut, she really looks like a completely different person."* These sentences are taken from a piece of writing that is most likely_____.
A. argumentative
B. expository
C. narrative
D. description
Clues: This describes the dressing style of Samrawit **Answer: D**

109. "In his eyes, I could see his desire to redeem himself, a need to hit me more times than I hit him, just for the sake of it." This text is taken from a piece of writing that is most _____.
A. expository
B. description
C. argumentative
D. narrative
Answer: A

110. "My coat was on the seat neat to me. I took my passport out of my pocket and put it in my bag in the luggage rack." These lines are taken from a piece of writing that is most likely_____.
A. argumentative
B. description
C. narrative
D. expository
Answer: C

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111. She crosses the road, trying to figure out how to navigate the town she has been away from for some fifteen years. This text is taken from a piece of writing that is most likely _____.
A. narrative
B. expository
C. argumentative
D. description
Answer: B

112. Which one of the following is **correctly punctuated**?
A. I have tried to sing dozens of times, she says, "but I cant."
B. "I have tried to sing dozens of times," she says "but I can't."
C. "I have tried to sing dozens of times" she says, "but I can't."
D. "I have tried to sing dozens of times," she says, "but I cant."
Answer: D

113. "Your responses will be held confidentially." Which of the following might be a good reason for writing the preceding sentence in the first few lines of a survey questionnaire?
A. Increasing response rate.
B. Appreciating respondents' participation
C. Limiting sample size
D. Requesting for permission
Answer: D

114. Which one if the following is **correctly punctuated**?
A. The meeting ended at dawn, noting had been decided.
B. The meeting ended at dawn noting had been decided.
C. The meeting ended at dawn; nothing had been decided.
D. The meeting ended at dawn. Nothing had been decided.
Answer: C

115. Which of the following sentences has correct capitalization?
A. Sara and Hana first met at a school called learning to Learn.
B. Sara and hana first met at a school called Learning to Learn.
C. Sara and Hana First met at a school called learning to Learn.
D. Sara and Hana first met at a School called Learning to Learn.
Answer: A

DIRECTION: Question 116-120 are based on the information in the table below. Read the questions carefully and answer them.

A survey was carried out to determine first year university students' essay writing strategies. The survey, in particular, focused on the students' language use strategies. Below is the summary of the data obtained in response to the questions asked.

Strategies	Percentages
Borrow phrases from books	15
Write in the local language (L1) and translate	30
Write directly in English	25
Outline in L1 and write in English	25
Other strategies	5

116. Which of the following is the likely title of the survey from which the data displayed in the above table were extracted?
A. A survey of first year university students' essay writing strategies.
B. A survey of essays written by first year university students.
C. Essay writing strategies as surveyed by first year university students.
D. Essay writing as surveyed by strategies of first year university students.
Answer: A

117. What is the most useful instrument used to gather the survey data displayed in the table?
A. Interview
B. observation
C. classroom Observation
D. Questionnaire
Answer: D

118. How many instances of use of L1 can be observed from the data reported in the table?
A. 1 B. 3 C. 2 D. 4
Answer: C

119. From the data in the above table, what proportion of first year university students considered in the survey wrote their essays directly in English?
A. half
B. three- fourths
C. two – thirds
D. a quarter
Answer: D

120. "Until future study finds a better solution, English teachers should work on how students can be helped to develop their outlines in English into essays." Which section of the above survey is the most likely appropriate place for the preceding statement?
A. Conclusion
B. Recommendation
C. Data analysis
D. Methodology
Answer: B

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111. She crosses the road, trying to figure out how to navigate the town she has been away from for some fifteen years. This text is taken from a piece of writing that is most likely _____.
A. narrative
B. expository
C. argumentative
D. description

Answer: B

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D. "I have tried to sing dozens of times," she says, "but I cant."

Answer: D

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D. Requesting for permission

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Answer: C

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Write directly in English	25
Outline in L1 and write in English	25
Other strategies	5

116. Which of the following is the likely title of the survey from which the data displayed in the above table were extracted?
A. A survey of first year university students' essay writing strategies.
B. A survey of essays written by first year university students.
C. Essay writing strategies as surveyed by first year university students.
D. Essay writing as surveyed by strategies of first year university students.

Answer: A

117. What is the most useful instrument used to gather the survey data displayed in the table?
A. Interview
B. observation
C. classroom Observation
D. Questionnaire

Answer: D

118. How many instances of use of L1 can be observed from the data reported in the table?
A. 1 B. 3 C. 2 D. 4

Answer: C

119. From the data in the above table, what proportion of first year university students considered in the survey wrote their essays directly in English?
A. half
B. three- fourths
C. two – thirds
D. a quarter

Answer: D

120. "Until future study finds a better solution, English teachers should work on how students can be helped to develop their outlines in English into essays." Which section of the above survey is the most likely appropriate place for the preceding statement?
A. Conclusion
B. Recommendation
C. Data analysis
D. Methodology

Answer: B

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111. She crosses the road, trying to figure out how to navigate the town she has been away from for some fifteen years. This text is taken from a piece of writing that is most likely _____.
A. narrative
B. expository
C. argumentative
D. description
Answer: B

112. Which one of the following is **correctly punctuated**?
A. I have tried to sing dozens of times, she says, "but I cant."
B. "I have tried to sing dozens of times," she says "but I can't."
C. "I have tried to sing dozens of times" she says, "but I can't."
D. "I have tried to sing dozens of times," she says, "but I cant."
Answer: D

113. "Your responses will be held confidentially." Which of the following might be a good reason for writing the preceding sentence in the first few lines of a survey questionnaire?
A. Increasing response rate.
B. Appreciating respondents' participation
C. Limiting sample size
D. Requesting for permission
Answer: D

114. Which one if the following is **correctly punctuated**?
A. The meeting ended at dawn, noting had been decided.
B. The meeting ended at dawn noting had been decided.
C. The meeting ended at dawn; nothing had been decided.
D. The meeting ended at dawn. Nothing had been decided.
Answer: C

115. Which of the following sentences has correct capitalization?
A. Sara and Hana first met at a school called learning to Learn.
B. Sara and hana first met at a school called Learning to Learn.
C. Sara and Hana First met at a school called learning to Learn.
D. Sara and Hana first met at a School called Learning to Learn.
Answer: A

DIRECTION: Question 116-120 are based on the information in the table below. Read the questions carefully and answer them.

A survey was carried out to determine first year university students' essay writing strategies. The survey, in particular, focused on the students' language use strategies. Below is the summary of the data obtained in response to the questions asked.

Strategies	Percentages
Borrow phrases from books	15
Write in the local language (L1) and translate	30
Write directly in English	25
Outline in L1 and write in English	25
Other strategies	5

116. Which of the following is the likely title of the survey from which the data displayed in the above table were extracted?
A. A survey of first year university students' essay writing strategies.
B. A survey of essays written by first year university students.
C. Essay writing strategies as surveyed by first year university students.
D. Essay writing as surveyed by strategies of first year university students.
Answer: A

117. What is the most useful instrument used to gather the survey data displayed in the table?
A. Interview
B. observation
C. classroom Observation
D. Questionnaire
Answer: D

118. How many instances of use of L1 can be observed from the data reported in the table?
A. 1 B. 3 C. 2 D. 4
Answer: C

119. From the data in the above table, what proportion of first year university students considered in the survey wrote their essays directly in English?
A. half
B. three- fourths
C. two – thirds
D. a quarter
Answer: D

120. "Until future study finds a better solution, English teachers should work on how students can be helped to develop their outlines in English into essays." Which section of the above survey is the most likely appropriate place for the preceding statement?
A. Conclusion
B. Recommendation
C. Data analysis
D. Methodology
Answer: B

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SECTION 7 – SCHOLASTIC APTITUDE

HOW TO WORK IT OUT

1. **Aptitude Test**

- **What is aptitude?** It is natural ability or skill at doing something. Its synonym is talent. (**Aptitude = talent**).
- **What is aptitude Test?** It is one designed to show whether somebody has the natural skill or ability for a particular job or course of education.
- **Aptitude Test** such as **Scholastic Aptitude Test** is a college university entrance examination that measure basic skills used in many different subject areas or **achievement tests** that measure knowledge of specific subject such as history or biology.
- **Scholastic Aptitude Test** measure skills that are gained over a period of many years. Consequently, it is impossible to study for them in the usual sense. In other words, you can't simply sit down a few days before these exams and learn everything that you need to know for them. Instead, you must begin now to develop the skills that these tests measure, including **vocabulary**; your **reading** and **computational skills;** your **ability** to think **reasonably** and **critically**; and your **understanding** of the basic principles of English.
- The **Scholastic Aptitude Test (SAT)** is divided into sections on **Verbal** and **Mathematical** ability.
- The types of English items included in **scholastic aptitude test** are called **VERBAL REASONIGN.**
- To prepare for this college entrance examination (SAT), familiarize yourself with the types of questions that the **verbal reasoning** contains.

☞ These include:

1. Analogy
2. Synonyms
3. Antonyms
4. Language usage
5. Grammatical meaning
6. Sentence completion
7. Reading comprehension
8. Logical reasoning
9. Analytical reasoning

Analogy

2. **Analogy** are words that are related to each other in some way. In an analogy question, you are given two words that are related, and you must determine what this relationship is. You then choose a second pair of words that are related in the same way.
- A typical analogy question looks like this.

Example: Analogies
SONG: REPERTOIRE

A. Score: Melody
B. Solo: Chorus
C. Instrument: artist
D. Suit: wardrobe
E. Benediction: church

- These are strategies to use in answering analogy questions.
1. First determine the relationship between containing **both of them.**
2. Then, find a second pair of words that you can substitute for the original pair.
   Several **suits** make up a **wardrobe.**
- Common relationships found in analogy questions include.

Types of analogy	Examples
Action to object	Play: clarinet
Cause to effect	Sun: sunburn
Item to category	Iguana: reptile
Object to its function	Pencil: writing
Object to its material	Curtains: cloth
Part to whole	Page: book
Time sequence	Recent: current
Word to antonym	Help: hinder
Word to synonym	Provisions: supplies
Worker and creation	Artist: sketch
Worker and tool	Lumberjack: saw
Type of characteristic	Dancer: agile

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1.2 Synonyms (SYN)

- **Synonyms** are words that have similar to meanings. Most **synonyms** don't mean exactly the same thing knowing the small differences in meaning between two synonyms enables you to each word precisely.
- Learning **synonyms** will help you broaden your vocabulary, expanding the range of words available to you and enabling you to express your ideas with precision.

Examples: Sincere, unaffected, heartfelt

- Show, display, exhibit, expose
- Expect, anticipate, hope, await
- Contain, hold, accommodate
- Rational, reasonable, sensible
- are synonym words

1.3 Antonyms

- An **antonym** is a word that is opposite in meaning to another word. **Antonym questions** provide a single word and ask you to select from a list of words the one that is most opposite meaning.
- A typical antonym question looks like this:

> **Example: Antonyms**
> **DEPRESS**:
> A. Force C. Clarify E. Loosen
> B. Allow D. Elate

- To answer an antonym question, use the following strategies.
  1. Look for a word that is opposite in meaning. Don't be thrown off by any synonyms – words that are similar in meaning – that included among the choices.
  2. Before looking at the choices, be sure that your know the meaning of the first word. Define it using any of the following methods:
     a. Think of another word or group of words that means the same thing.
     b. Think of a sentence or sentences that use the word, then try to arrive at an exact definition.
     c. Try analyzing the parts of the word.
  3. Once you know the meaning of the first word, look at the choices. If none of these is obviously the correct antonym, try either or both of the following strategies.

     a. Eliminate any odiously correct answers.
     b. Remember that many words have more than one meaning. If none of the choices seems to be opposite in meaning, think of other meanings of the first word.

Therefore, the answer to the sample question is (D).

1.4 Language usage questions (usage questions)

- This kind of question can ask either the corresponding meaning or phrases or idioms or the grammatical error within the same sentence.
- **Idioms (especial meaning phrases)** can be done through the contextual meaning they have in a sentence or we need to know the especial expressions they originally have.
- **Grammatical usage (seeking error)** question require that you recognize wiring that is not standard written English. In usage questions, four words or phrases are underlined and lettered. You must either choose the underlined part the contains an error or mark the answer corresponding to no error.
- Here is a typical usage question.

> **Example: Usage questions**
> He spoke blunty and angrily to we spectators. No

- Use the following strategies to answer usage questions.
  1. Read the sentences in its entirety.
  2. Look for errors of the following kinds.
     a. Incorrect word error
     b. Errors in punctuation
     c. Errors in capitalization
     d. Errors in grammar
     e. Improper, awkward, or unclear sentence structure
  3. Common grammatical errors found in usage questions include the following: incorrect verb tenses, improper agreement of pronoun and antecedent or of subject and verb, improper form of the pronoun, and use of an adjective for an adverb or an adverb for an adjective. Common errors in sentence structure include **fragments, dangling modifier, improper parallelism, and run – on caused by comma faults.**
  4. Remember that the error, if there is one, appears in one of the underlined sections of the sentence.

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- **Thus,** the correct answer to the example question is (c). The object pronoun '**us**' should follow the preposition.

1.5 Grammatical Meaning questions

This is the same as/ to mean language usage. This type of question is to ask you to choose a sentence that carries the closest/ nearest meaning to the given sentence (with original sentence).

1.6 Sentence completion questions

- **Sentence completion** questions measure both your vocabulary and your ability to recognize relationships between parts of a sentence. You are given a sentence from which one or two words have been removed. You must choose words that best fit the meaning of the rest of the sentence.
- A typical sentence completion question looks like this.

Example: Sentence completion

The author was determined to _____ his own conclusion, so he _____ any information that didn't support it.

A.	Unhold . . . ignored	D.	disprove . . . destroyed
B.	Revise . . . destroyed	E.	reverse . . . confiscated
C.	Advance . . . devised		

- To answer sentence completion questions, use the following strategies.
  1. Read the incomplete sentence carefully, noting any key words. Pay particular attention to words that indicate contrast (**but however, on the to her hand**) or similarly (**and, another, the same as**).
  2. Try each of the choices in the sentence, eliminating any that make no sense or that are grammatically incorrect.
  3. Don't misled by answers that contain only one word that fits well into the sentence. Incorrect answers often contain one word that makes sense and another one that doesn't. In the example, the first word in answer c makes sense, but the second word doesn't. **Answer A is correct.**
  4. Look for grammatical clues in the sentence. Does the sentence call for a verb, an adverb, a noun? If a verb is required, what should it tense be? By asking such questions, you can sometimes eliminate incorrect answers.

1.7 Reading comprehension question

- The **reading comprehension** questions measure your ability to understand various kinds of reading passages. The passages are taken from such fields as **biological and physical sciences, the humanities** and **the social science.**
- To answer **reading comprehension** questions, use the following strategies.

1. Expect questions on the following:
   a. The central thought of the passage.
   b. Specific details in the passage.
   c. Conclusions that can be made based upon information provided in the passage.
   d. The meaning of words in the passage.
   e. The mood of the passage.
   f. Specific techniques used by the writer of the passage.
   g. Relationships between things in the passage.
2. As you read the passage, identify the main idea and pay particular attention to transitional and other words that show relationships.
3. Read all the choices before you select an answer.
4. Answer the questions based on material in the passage, not on your own knowledge or opinions.
5. Don't choose an answer simply because it is true. Make certain that you answer the questions being asked.
6. Don't be misled by answers that are partially correct.
7. Budget your time. Set a time limit for reading each passage and answering questions on it. Don't waste time on impossibly difficult questions. Refer to your watch to make sure that you are not spending too much time.

1.8 Logical reasoning question

- Logic is the study of **reasoning (logical reasoning).**
- One of the most basic types of **reasoning** is known as **induction.**

A. Inductive reasoning

- This is the process arriving at a general conclusion based up on **specific facts.**

A simple example of **induction** may be illustrated as follows.

1. First, you observe several specific facts. In the past, candidate smith has always voted in favor of federally – funded jobs programs. In a speech before the league of women voters, candidate smith expressed his support for federally – funded any

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care centers. In a new conference, candidate – smith stated that he plans to introduce legislation that will provide federal assistance to pay the utility bills of people who need help.

2. Then, having noticed a similarity between these specific facts, you draw a conclusion.

☞ Candidate smith usually supports federally – funded social programs.

The conclusion that is drawn through inductive reasoning is called **generalization**. The group of specific on which the generalization is based is called a **sample**.

☞ [specific fact + specific fact → generalization]

B. Deductive reasoning

▪ In **deductive**, you start with a generalization and then derive a specific fact from it.

[Generalization + related fact → specific fact]

E.g.1 Snakes that have fangs are poisonous (generalization). The asp viper has fangs (**related facts**). Therefore, the asp viper is poisonous (specific fact).

Example 2: All ballads are poems (**Major premise**)
"Barbara Allen" is a ballad. (**Minor premise**)
Therefore, "Barbara Allen" is a poem. (**Conclusion**)

1.9 Analytical Reasoning questions

▪ **Analytical reasoning** is logic that is inferred through the virtue of the statement's own content, and it can be used to solve many types of problems in almost any discipline. This analytical aptitude test assesses inductive and deductive reasoning skills.

Antonyms (1 – 2)

DIRECTION: Choose the word or phrase that is most nearly **opposite** in meaning to the word in capital letters and **blacken** the letter of your choice on the separate answer sheet.

Antonym: An antonym is a word that means the **opposite** of another. Dictionary lists **antonyms** for many words.

Examples: abundant: scarce ban: allow
Negative: positive drab: colorful
Precise: in exact exterior: interior
Descendant: ancestor effect: cause
Temporary: permanent meak: bold

1. IMPROMPTU
A. Natural C. Planned
B. Unrehearsed D. Spontaneous

Answer: C

Clue: importantu (adj) done without being planned or rehearsed.

2. ALIEN
A. Inimical C. Parochial
B. Affluent D. Indigenous **Answer: D**

Synonyms (3 – 4)

Direction: choose the word that is most nearly **similar** in meaning to the word in capital letters and **blacken** the letter of your choice on the separate answer sheet.

Example: abundant: plentiful impose: force
foretell: predict walked: padded
obligation: duty minimum: least

3. PARTISAN
A. Biased C. Neutral
B. Objective D. Impartial

***Clue*: PARTISAN (adj):** someone who is **Partisan** strongly supports a particular person or cause, often without thinking carefully about the matter. **Answer: A**

4. NEUTRAL
A. Unkind C. Mean
B. Precious D. Indifferent

***Clue*: NEUTRAL (adj):** If you say that something **neutral**, you mean that they lacks noticeable or strong qualities . **Answer: D**

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Analogy (5 – 9)

DIRECTIONS: Answer questions **5 – 6** based on the analogy used to show relationship between the first two words in capital letters. Answer questions **7 – 9** based on the pair that best expresses a relationship **similar** to that expressed in the original capitalized pair. Then, find the corresponding question number on the separate answer sheet and **blacken the letter of your choice.**

Analogy (5 – 9)

Vocabulary is often a measure of how much a person knows for this reason your knowledge of words will be tasted several times throughout your years in school. There for many types of vocabulary test. One type of vocabulary test asks you to identify relationships between **pairs of words.**

This relationships are called **analogies.** The following example is an item from a test of analogies.

DRAB: DULL

a. Mealth: poverty
b. Joy: sorry
c. Wisdom: intelligence

Clue:

The first step in answering this test item is to identify the relationship between the two words in **capital letters.**

In this item the words are **synonyms,** since **drab,** and **dull** have similar meanings. The next step is to find the other pair of words that have the same relationship. **Answer: C**

LATTER: FORMER

A. old: ancient
B. handsome: attractive
C. close: open

Clue: The first two pairs are both **antonyms.** **Answer: C**

5. LAW to ANARCHY as MEDICINE is to _____

A. illness C. health
B. recovery D. death

Clue: **ANARCHY:** If you describe a situation as **anarchy,** you mean that nobody seems to be paying any to **RULES** or **LAWS.**

Answer: A

6. UNCLE is to NEPHEW is AUNT is to _____

A. cousin C. sister
B. daughter D. niece

Clue: **NEPHEW:** Is your sister's or brother's **son** (you're an uncle for him) is male

NIECE: Is your sister or brother's **daughter** (you're an aunt for him) **is** male

Answer: D

7. ETYMOLOGY: DERIVATION

A. ecology: urbanization
B. autonomy: authorization
C. taxonomy: classification
D. economy: rationalization

Clue: **ETYMOLOGY:** is the study of the origins historical development of words.

DERIVATION: words origins /source

TAXONOMY: Is the process of naming and classifying things such as animals and plants into groups within a large system according to their similarities and differences

Answer: C

8. TONE: SCALE

A. noise: wave C. waves: sound
B. colour: spectrum D. verse: rhyme

Clue: **TONE:** Is a difference in pitch between two musical notes equal to two semitones.

SCALE: In music, scale is a fixed sequence of musical notes, each one higher than the next, which begins at a particular not. **Answer: D**

9. WORD: SENTENCE

A. picture: pencil C. note: bar
B. writing: paper D. word: letter

Clue: **WORD** makes a sentence (a sentence is a groups of related word)

WORD is made of **letters.** (a groups of letters make a word) **Answer: D**

Language Usage (10 – 15)

DIRECTIONS: For questions **10 – 14** choose the one that most nearly has the same meaning as the underlined word/words from the given alternatives and **blacken** the letter of your choice on the separate answer sheet.

10. The police have made headway with their investigation in Friday's murder.

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A. finding
B. allegation
C. direction
D. progress

Clue: **headway (idiom)**: If you make a **headway**, you **progress** towards achieving something.

Answer: D

11. The father stood his ground in support of his son despite several complaints from friends.
A. maintained his position
B. rejected his position
C. was reluctantly
D. changed his position

Clue: **Stand group /hold ground:** If you stand your ground or hold your ground, you **continue** to support a particular opinion when other people are opposing you or trying to make you change your mind.

Answer: A

12. Not being aware of the legal implications of his action, an arrest was made by the police officer.
A. The police officer violated the law.
B. The arrested person was not an offender intentionally
C. The police officer called the lawyer to arrest the offender
D. The police officer made an arrest with no clear understanding of the consequence.

Clue: The other way of saying /expressing the underlined clause is: The person was arrested as an offender as a result of the police officer's misunderstanding of the legal implications.

Answer: D

13. The computer system was down all morning and afternoon yesterday.
A. depressed
B. well functioning
C. degraded
D. not working

Answer: D

14. Being an Ethiopian, I felt like a fish out of water in the group of Russians.
A. misplaced
B. uncomfortable
C. annoyed
D. abandoned

Clue: The right place given to fish is in the water. Fish out of water means death. At the same time if we go somewhere strange, we fell something like this (This is an idiom)

Answer:A

15. Below are two underlined sentences followed by a question. Red the sentences, then choose the best answer to the question.
Knowledge of another language fosters greater awareness of cultural diversity among the peoples of the world.
Individuals who have foreign language skills can appreciate more readily other peoples values and ways of life.
How are the two sentences related?
A. They establish a contrast.
B. They repeat the same idea.
C. They contradict each other.
D. They present problems and solutions

Clue: The second sentences repeat or give more emphasis on the first sentences (nothing is new).

Answer: B

DIRECTION: Each sentence below has one or two blanks. Each blank indicates that something has been omitted. Beneath the sentence are four sets of words. Choose a word or the set of words that best fits the meaning of the sentence as a whole and **blacken** the letter of your choice on the separate answer sheet.

Sentence completion (16 – 20)

Sentence completion questions measure both your vocabulary and your ability to recognize relationships between parts of a sentence. You are given a sentence from which one or two words have been removed. You must choose words that best fit the meaning of the rest of the sentence. So read the incomplete sentence carefully, noting any key words. Pay particular attention to words that indicate contrast(**but, however, on the other hand**) or similarity(**and, another, the same as**)

Example: The author was determined to ____ hisown conclusion, so he ___ any information that didn't support it
A. Uphold /ignored
B. Revise /destroyed
C. Advance/ devised
D. disprove/ distorted
E. reverse /confiscated

Answer A

16. The young graduate was quickly promoted when his boss saw how _____ he was.
A. tired
B. hasty
C. destitute
D. diligent

Clue: Why was the young graduate quickly promoted? Because his boss saw him **hardworking /clever/ diligent**

Answer: D

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17. Ever since I arrived at the school last Thursday, I have been _____ by the rude behavior of the students. In contrast, In contrast, the teachers were _____.
A. shocked; surprised
B. stunned: indifferent
C. surprised; flabbergasted
D. neutral; dispassionate

Clue: the rude behavior of someone always makes some body **upset /shocked/ annoyed**, so the speaker in this question as also been shocked. The phrase **'in contrast'** on the other hand tells us the opposite thing that is the teachers were **surprised**.

Answer: A

18. Although she did not consider herself _____, she felt that the inconsistencies in his story _____ a certain degree of incredibility on her part.
A. a charlatan; dignified
B. an opposite; justified
C. a skeptic; warranted
D. a hypocrite; demonstrated

Answer: D

19. The deadline for the project is fast _____; we are doing everything we can to complete it on time.
A. outgoing
B. arriving
C. drawing
D. soaring

Answer: B

20. Most people attempt to present a particular image of themselves others: they also _____ a lot of time and energy in developing such an image:
A. want
B. ratio
C. loss
D. invest

Clue: If you **invest** time or energy in something, you spend a lot of time or energyon something that you consider to be useful or likely to be successful **'loss'** is noun not verb. **Answer: D**

Reading Comprehension (21 – 29)

DIRECTIONS: Each question from **21-29** is based on the information stated or implied in each passage. Read the passage carefully and then find the corresponding question number on the separate answer sheet and blacken the letter of your choice.

Reading Passage 1

The restaurant opens at 6:00 a.m. to serve breakfast which must be ordered by 9:00 a.m. Lunch is served between 11:30 a.m and 2:30 p.m. Dinner is served between 6:30 p.m. and 8:30 p.m. Guests can be accommodated at lunchtimes and dinnertimes provided that 24 hours notice has been given. Vegetarian options are always available, but vegans should notify the catering coordinate ahead of time as should anyone with special dietary requirements. This includes nut, gluten and soybean allergies, etc.

21. Which of the following statements is implied in the passage?
A. Guests cannot be accommodated at breakfast.
B. Guests can order lunch at 9:45 a.m. if they wish.
C. Vegetarian should notify the catering coordinator.
D. Someone allergic to eggs should notify the catering coordinator.

Answer: B

22. The intended audience for this passage is most likely
A. tourists
B. special guests of the restaurant.
C. all customers of the restaurant.
D. service provides of the restaurant.

Answer: B

23. Which of the following is **NOT** implied in the passage?
A. Guests cannot expect lunch to bee served at 3:00p.m.
B. 12:00 a.m. is too early to arrive at the restaurant for lunch.
C. Guests that come at 10:00 a.m. for breakfast are not served.
D. Guests cannot be accommodated for lunch unless they give one day notice ahead of time.

Answer: D

24. The main suppose of his passage is to explain about the
A. service provisions.
B. timing of services
C. requirements of guests.
D. types of foods available.

Answer: A

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Reading Passage 2

Before AbebeBikila, there were no runners in Africa, or at least no runners that the world knew about. After Bikila won Olympic marathon in Rome in 1960, we Africans all started thinking; "Look – he is one of us. If he can do it, we can do the same."

Let me explain it in a different way. Before the Wright brothers **invented** the first successful airplane, no humans could fly, and most people didn't believe such a thing was possible – yet now we're sending rockets to Mars and **who knows where next**. In Africa before Bikila, there were no successful runners, and most of us didn't believe there ever would be. Now there are thousands of us **winning races** all over the world, **setting new standards** and **breaking records.**

In one my offices there is a positer that shows all the great champions Ethiopia has produced. Bikila is there right at the center. He is an **icon** for the whole of Africa, a personal **hero** of mine. Because of AbebeBikila, I am trying to win an Olympic marathon, just like him.

(Contributed by Haile Gebreselassie, TIME, November 13,2006, vol. 168, No.21, p, 108)

25. Olympic marathon is to AbebeBikila as _____ is to the Wright brothers.
    A. airplane
    B. rocket to mars
    C. invention
    D. successful runner

    ***Clue*:** Wright brothers were the first two brothers to make invent the first flying aircraft. **Answer: A**

26. According to the passage, who set new records in the Olympics Marathon?
    A. Haile Gebreselassie
    B. BelaynehDinsamo
    C. African Olympics Marathon runners
    D. Olympics Marathon runners after AbabeBikila **Answer: A**

27. Which of the following statements is a valid inference from the passage?
    I. Haile Gebreselassie, just like AbabeBikila, is an icon for the whole of Africa.
    II. AbebeBikila is Haile Gebreselassie's hero.
    III. After AbebeBikila, there are now thousands of runners in Africa; yet only a few have won races.

    A. I only B. II only C. III only D. II and III

    Answer: B

28. In the quotation, "… If the can do it, we can do the same" (paragraph 1), "he" and "we" refer to _____ and _____, respectively
    A. AbabeBikila: Africans
    B. Africans: AbebeBikila
    C. AbebeBikila: the Wright Brothers
    D. The Great Champions: Ethiopians **Answer: A**

29. The term "icon" (paragraph 3) could mean all of the following **EXCEPT**_____
    A. idol
    B. image
    C. symbol
    D. runner

    ***Clue*: icon (N):** If you describe something or someone as an icon, you mean that they are important as a **symbol, idol, image.**

    Answer:D

Logical and Analytical Reasoning (30 – 35)

DIRECTIONS: In this section, each question or a group of questions is based on a passage or set of conditions. Read each question carefully and select the best answer and **blacken** the letter of your choice on the separate answer sheet.

30. All good football teams want to win, and all players of a footballteam who want to win eat a well-balanced diet. Therefore, all players of a football team who do not eat a well-balanced diet are bad players.

    If the assumptions of the argument above are true, then which of the following statements must be true?
    A. All football teams who want to win are good teams.
    B. Some good football players do not eat a well0balanced diet.
    C. No football player who does not eat a well-balanced diet is a good player.
    D. Every football player who eats a well-balanced diet is a good player.

    ***Clue*:** All players of a football team who don't eat a well-balanced diet are bad player implies that every football player who eats a well-balanced diet is a good player. **Answer:D**

31. Abera is older than Zeinebe, but he is either younger than or as old as Meron. Which one of the following statement is **TURE** about the relationship between the ages of Zeineba and Meron?
    A. Meron is as old as Zeineba
    B. Meron is younger than Zeineba.

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C. Zineba is younger than Meron.
D. The relathionship cannot be determined.

Clue : To understand clearly, let's use this label.
Abera is **older than** Zenebe.
Let say the age of Abera is 20 and Zenebe's is **15**.
But Abera is either**younger than** or **as old as** Meron.
Let say Meron's age is **25** or **20**.
So the conclusion should be:
Meronis older thanZenebe or
Zenebeis younger thanMeron

Answer: C

32. Given the following two premises, which one of the following conclusions can follow?
Premise I: All hasty generalizations are erroneous.
Premise II: No erroneous generalization is acceptable.
A. No hasty generalization is acceptable.
B. All hasty generalization are acceptable.
C. All erroneous generalizations are hastily made.
D. No hastily made generalization is unacceptable.

***Clue*: Premise**: Is something that you **suppose** is **true** and that you use as a basis for developing an idea.

Examples

Major premise: Some mammals lay eggs.
Minor Premise: The horse is a mammal.
Conclusion: Therefore, horses lay eggs.

Both the major and minor premises of this argument are true some mammals, the platypus and the echidna, for examples, do lay eggs. It is also true that the horse is a mammal. However, the conclusion of this argument is false because the argument has an invalid.

All teenagers love loud music
Rolanda is a teenager.
Therefore, Rolanda loves loud music

Answer: A

33. The cost of manufacturing motor bicycles in china is fifteen percent less than the cost of manufacturing motor bicycles in Ethiopia. Even after transportation fees and import taxes are added, it is still cheaper to import motor bicycles form China to Ethiopia than no produce motor bicycles in Ethiopia.

Which one of the following assertions is best supported by this information?
A. Labor costs in china are fifteen percent below those in Ethiopia.
B. Importing motor bicycles into Ethiopia will eliminate fifteen percent of the manufacturing jobs in Ethiopia.
C. The import taxes on a motor bicycle imported from china to Ethiopia is less than fifteen percent of the cost of manufacturing the motor bicycle in Ethiopia.
D. The costs of transporting a motor bicycle from China to Ethiopia is more than fifteen percent of the cost of manufacturing the motor bicycle in china.

Answer: C

34. If you present a red pass, then you may enter the compound.
If the statement above is true, which of the following must also be true?
I. If you do not present a red pass, then you may not enter the compound.
II. If you may enter the compound, then you must have presented a red pass.
III. If you may not enter the compound, then you did not present a red pass.
A. I only
B. III only
C. I and II only
D. II and III only

Answer: A

35. There are 850 bottles to be filled: Abem and Nati working independently, but at the same time take 30 minutes to fill the bottles.
Statement 1: Nati fills half as Amny bottles as Abem
Statement 2: Abem would take 45 minutes by himself.
Which of the statements above make it possible to answer the question, "How long should it take Nati working by himself to fill the bottles?"
A. Each statement along is sufficient.
B. Statements 1 and 2 together are not sufficient.
C. Statement 2 alone is sufficient, but statement 1 alone is not sufficient.
D. Statement 2 alone is sufficient, but statement 1 alone is not sufficient.

Answer: D

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1. If the product of three consecutive positive integers is equal to their sum, what is the largest integer?
A. 3 B. 2 C. 1 D. 0

Solution: Let $2n-1$, $2n$ and $2n+1$, are consecutive positive integer for $n = 1, 2, 3, 4\ldots$

$(2n-1)(2n)(2n+1) = (2n-1) + 2n + (2n+1)$

$\Rightarrow 8n^3 - 2n = 6n$

$\Rightarrow 8n^3 - 8n = 0 \Leftrightarrow 8n(n^2-1) = 0$

$\therefore n = 0$ and $n^2 - 1 = 0$, that is $n = 1$, or $n = -1$

However, we take only $n = 1$

Therefore 1, 2 and 3 are the number.

Answer: A

2. What is the missing value of b in the sequence 33, b, 19, 12, 5?
A. 25 B. 26 C. 27 D. 29

Answer: B

3. If $a + 3b = 13$ and $3a + 3b = 15$, what is the arithmetic mean of a and b?
A. 3.5 B. 3 C. 2.5 D. 2

Solution: we have $3a + 3b = 15 \Rightarrow a + b = 5$

$(a + 3b) - (a + b) = 13 - 5$

$2b = 8 \Rightarrow b = 4$ and $a = 1$

$$\therefore A.M = \frac{a+b}{2} = \frac{1+4}{2} = 2.5$$

Answer: C

4. If $7x + 17 = 49$, what is the value of $\sqrt{7x+49}$?
A. 7 B. $4\sqrt{7}$ C. $8\sqrt{5}$ D. 9

Solution:

$7x + 17 = 49$ equivalent to

$7x + 17 + 32 = 49 + 32 \leftarrow$ Adding 32 to both side

$7x + 49 = 81$

$\therefore \sqrt{7x+49} = \sqrt{81} = 9$

Answer: D

5. If $x = 16 \times 7$, which one of the following is NOT an integer?
A. $x \div 2$ C. $x \div 4$
B. $x \div 3$ D. $x \div 7$

Solution: $x = 16 \times 7 = 112$, therefore,
112 is visible by 2, 4 and 7

Answer: B

6. If the product of four integers is negative, how many of the integers can be negative at most?
A. 1 B. 2 C. 3 D. 4

Solution:

(+ve)(+ve)(+ve)(−ve) = −ve ← one negative and 3 positive.

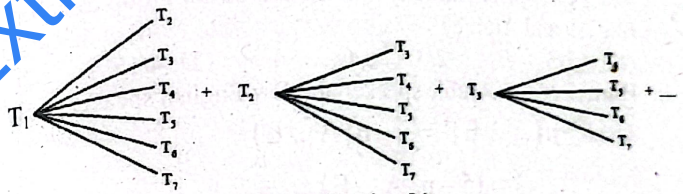
(+ve)(−ve)(−ve)(−ve) = −ve ← 3 negative and 1 positive.

Answer: C

7. A team of seven teachers met for a discussion. Each teacher shook hands once with each other. Howe many handshakes took place?
A. 36 B. 28 C. 21 D. 15

Explanation:

Let each of them are $T_1, T_2, T_3, T_4, T_5, T_6, T_7$



$6 + 5 + 4 + 3 + 2 + 1 = 21,$

Answer: C

8. If x and y are integers such that $2x - 4 > 0$ and $2y + 4 < 0$, then which of the following must be true?
A. xy is negative C. xy is odd
B. xy is positve D. xy is even

Solution: Here

$\rightarrow 2x - 4 > 0 \Rightarrow x > 2 = \{3, 4, 5, \ldots\}$

$\rightarrow 2y + 4 < 0 \Rightarrow y < -2 = \{-6, -5, -4, \ldots\}$

$\therefore$ xy is negative,

Answer: A

9. What is the value of $\sqrt{2} \times \sqrt{15}$?
A. 15 C. $\sqrt{30}$
B. 30 D. $\sqrt{17}$

Solution: $\sqrt{2} \times \sqrt{15} = \sqrt{2 \times 15} = \sqrt{30}$

Answer: C

10. If a is 9 more than b, b is 4 more than c, and d is 3 more than c, then what is the positive difference between a and d?
A. 13 B. 12 C. 11 D. 10

Hint: $a = 9 + b$, $b = 4 + c$, and $d = 3 + c$

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$|a-d| = |9+b-(3+c)| = |9-3+b-c| = |6+b-c|$

But we have $b = 4 + c$

$\therefore |6+4+c-c| = |10| = 10,$

11. One ordering of the letters A , B, C and D from left to right is BACD. What is the total number of orderings of these letters from left to right, including BACD?

Answer: D

A. 12 B. 16 C. 20 D. 24

Solution: $4P_4 = \frac{4!}{(4-4)} = \frac{4!}{0!} = 4! = 4\times3\times2\times1 = 24$

12. In a group of 15 persons, 7 can speak Arabic, 8 can speak English, and 3 can speak neither, what friction of the group can speak both English and Arabic?

Answer: D

A. 1/3 B. 2/3 C. 1/5 D. 4/15

Hint: Let A = Arabic speaker and E = English speaker

Recall: $n(A\cup E)' = \cup - n(A\cup E)$

$3 = 15 - n(A\cup E)$

$\Rightarrow n(A\cup E) = 15-3 = 12$

We know that $n(A)+n(E) = n(A\cap E)+n(A\cup E)$

$7+8 = n(A\cap E)+12$

$\therefore n(A\cap E) = 3 \leftarrow$ both speakers

$\therefore$ From group can speak both language is fraction of $\frac{3}{15} = \frac{1}{5}$

13. Sixty bottle contain a total of 24 liters of mango juice. If each bottle contains the same amount of mango juice, how many liters of juice are in each bottle?

Answer: C

A. 0.2 B. 0.3 C. 0.4 D. 0.8

Hint: $60x = 24 \Rightarrow x = \frac{24}{60} = \frac{2}{5} = 0.4$

14. Halima wants to tile the floor of her bedroom. The floor is rectangular and measures 12 meter by 8 meter. If it costs 2.50 birr per square meter for the materials, then what is the total cost of the materials in Birr for tiling the bedroom?

Answer: C

A. Birr 240 C. Birr 200

B. Birr 220 D. Birr 160

Explanation: Area = $12m \times 8m = 96m^2$

$$\left.\begin{array}{l} 1m^2 = 2.5Birr \\ 96m^2 = x \end{array}\right\} \Rightarrow x = 96\times2.5 = 240Birr$$

Answer: A

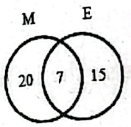
15. There are 27 students in a mathematic class and 22 students in an English class seven of these students take mathematics and English. What is the ratio of the number of students taking only English to those taking mathematics?

A. 2:5 C. 4:3

B. 3:4 D. 7:6

Solution: Let M = student in mathematics class

E = student in English class



✓ Number of student taking math's only = 20

✓ Number of student taking English only = 15

$\text{Ratio} = \frac{15}{20} = 3:4$ **Answer: B**

16. Compare the following two quantities.

Quantity A	Quantity B
$11\times\sqrt{2}$	$9\times\sqrt{3}$

A. The two quantities are equal.

B. Quantity A is greater than quantity B

C. Quantity B is greater than quantity A

D. The two quantities cannot be compared.

Solution: comparing means ordering, <, >, =

- Quantity A = $11\times\sqrt{2} = \sqrt{121}\times\sqrt{2} = \sqrt{121\times2} = \sqrt{242}$
- Quantity B = $9\times\sqrt{3} = \sqrt{81}\times\sqrt{3} = \sqrt{81\times3} = \sqrt{243}$

Quantity B > Quantity A **Answer: C**

17. Suppose that $a < b < c < 0$ and compare the following two quantities.

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Quantity A	Quantity B
$\frac{c}{b}$	$\frac{c}{a}$

A. The two quantities are equal
B. Quantity B is greater than quantity A
C. Quantity A is greater than quantity B
D. The two quantities cannot be compared.

Hint: let a = -4, b = -3, c = -2 Quantity

A : $\frac{c}{b}=\frac{-2}{-3}=\frac{2}{3}$ are quantity B : $\frac{c}{a}=\frac{2}{4}\Rightarrow\frac{2}{3}>\frac{2}{4}$ **Answer: C**

18. Suppose b and c are any real numbers?

Quantity A	Quantity B
b = c – 1	c = b + 1

A. The two quantities are equal
B. Quantity A is less than quantity B
C. Quantity A is greater than quantity B
D. The relationship cannot be determined from the information given.

Hint: Quantity (A) : b = c – 1 ⇒ c = b + 1
Quantity B : C = b + 1 **Answer: A**

19. Consider the following two quantities.

Quantity A	Quantity B
The number of prime numbers that are divisible by 3	The number of prime numbers that are divisible by 5

A. The two quantities are equal
B. Quantity A is less than quantity B
C. Quantity A is greater than quantity B
D. The relationship cannot be determined from the information given.

Hint: The only prime number that are:
Divisible by 3 is 3 itself
Divisible by 5 is 5 itself **Answer: A**

Questions 20 – 23 below are based on the hypothetical data given in Table 1.

Table 1. Family size, annual income and annual expenses (in Ethiopian birr, ETB) of 10 families in Addis Ababa

Code	Family size	Annual income (in ETB)	Annual expenses (in ETB)
Family 1	5	120,000	95,000
Family 2	3	80,000	55,000
Family 3	2	75,000	72,000
Family 4	5	150,000	110,000
Family 5	6	148,000	120,000
Family 6	8	90,000	70,000
Family 7	4	84,000	60,000
Family 8	6	90,000	60,000
Family 9	7	72,000	66,000
Family 10	7	84,000	72,000
Sum	**60**	**993,000**	**780,000**

20. For which three of the families are the ratio of annual income to family size the largest?
A. Families 3, 4 and 5
B. Families 1, 2 and 4
C. Families 4, 2 and 5
D. Families 2, 5 and 7
Answer: C

21. Which one of the following can be inferred from the data in table 1 above?
I. The mean annual saving of the ten families in ETB 21, 300
II. The mean family size of the ten families is 6
III. The mean annual expense of the ten families is ETB 99,330.
A. I and II
B. II and III
C. I and III
D. I, II and III

Solution: Heat at:

I. mean annual saving = $\frac{993,000-780,000}{10}=21,300$

II. mean family size = $\frac{60}{10}=6$

III. mean annual exp = $\frac{780,000}{10}=78,000$ **Answer: A**

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22. If the difference between each family's annual income and annual expenses directly goes into their bank saving accounts, which two families have the lowest and highest annual savings, respectively?
A. Families 3 and 1 C. Families 9 and 10
B. Families 9 and 4 D. Families 3 and 4 **Answer: D**

23. For which one of the families is the ratio of annual income3 to annual expenses closest to 1,00?
A. Family 10 C. Family 9
B. Family 3 D. Family 5 **Answer: C**

Questions 24 – 25 below are based on the data in the pie chart.

Annual Expenditure

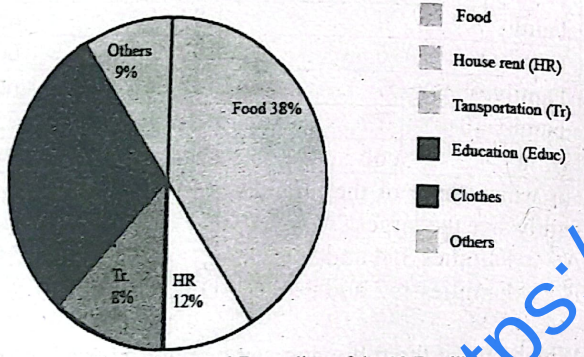

Figure 1. Annual Expenditre of the 10 Families

24. From the pie chart, what is the ratio of the annual expenditure on food to the annual expenditure on education?
A. 9:18 B. 19:9 C. 2:1 D. 3:2
Solution: 38% : 18% = 19:9 **Answer: B**

25. Considering the data provided in table 1 and figure 1 above, which one of the following can be inferred?
I. The 10 families spend ETB 296,400 annually on food.
II. The ratio of annual expenditure on transportation to that on clothing is 8:15
III. The ratio of annual expenditure on house rent to the corresponding expenditure on clothing is 4:5.
A. I only C. II only
B. III only D. I, II and III
Solution: Here at
I. 38% of 780,000 = 0.38 × 780,000 = 296,400
II. 8% : 15% = 0.08: 0.15 = 8:15
III. 12%: 15% = 12:15 = 4:5 **Answer: D**

Section II: Quantitative reasoning (Questions 1 – 24)

DIRECTIONS: Solve each of the problems given in questions **1 – 21** below and select the best answer choice from the given alternatives. Then, find the correct question number and **blacken** the letter of your choice on the separate answer sheet provided.

1. What decimal of an hour is a second?
A. 0.0167 B. 0.000126 C. 0.0256 D. 0.00027
Hint:

$$1\text{sec} = \frac{1}{3600}\text{hour} = 0.00027$$

Answer: D

2. What least number must be added to 1056, so that the sum is completely divisible by 23?
A. 18 B. 2 C. 3 D. 21
Hint: $46 \times 23 = 1058 - 1056 = 2$. **Answer: B**

3. The largest 4 digit number exactly divisible by 88 is:
A. 9768 B. 9988 C. 9944 D. 8888
Hint: $10{,}000 = (113)(88) + 46$ **Answer: C**

4. If $n = 9 \times 28 \times 26$, which of the following is **NOT** an integer?
A. n/18 B. n/32 C. n/21 D. n/36
Hint: $n = 9 \times 28 \times 26 = 3^2 \times 2^3 \times 7 \times 13$.

A) $\frac{n}{18} = \frac{3^2 \times 2^3 \times 7 \times 13}{2 \times 3^2} = 14 \times 26 \leftarrow$ integer

B) $\frac{n}{32} = \frac{3^2 \times 2^3 \times 7 \times 13}{2^5} = \frac{3^2 \times 7 \times 13}{2^5} \leftarrow$ not integer

Answer: B

5. The sum of the first nine natural numbers (1, 2, …,9) is
A. 90 B. 55 C. 45 D. 100
Hint: sum = (average) (number of terms)

$$= \left(\frac{1+9}{2}\right)(9) = 45$$

Answer: C

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6. If $a + b = 10$ and $a - b = 5$, then $a^2 - b^2$ is equal to

A. 50 B. 1/2 C. 5 D. 2

Hint: $a^2 - b^2 = (a - b)(a + b) = (10)(5) = 50$

Answer: A

7. In which of the following pair of number is the first number or expression necessarily less than the second number or expression?

A. 3/7 ; 9/49
B. x/4 ; 14/x, for x > 0
C. $(2/3)^2$; $2/3^2$
D. [(3/7)(24/25)] (7/3); 1

Hint: $\left(\frac{3}{7}\right)\left(\frac{7}{3}\right)\left(\frac{24}{25}\right) = \frac{24}{25} < 1$

Answer: D

8. Which one of the following fractions is greater than ¾ and less than 5/6?

A. 1/2 B. 4/5 C. 2/3 D. 9/10

Hint: $\frac{3}{4} < \frac{4}{5} < \frac{5}{6}$

Answer: B

9. $(100 \times 5^4) = ?$

A. 62500 B. 76600 C. 67000 D. 77200

Hint: $(100 \times 5^4) = (100 \times 625) = 62500$

Answer: A

10. $(935421 \times 625) = ?$

A. 575648123
B. 584649124
C. 584638125
D. 935425

Hint: 935421 × 625 = 584638125

Answer: C

11. In a certain school, 20% of students are below 8 years of age. the number of students above 8 years of age is 2/3 of the number of students of 8 years of age which is 48. What is the total number of students in the school?

A. 72 B. 150 C. 80 D. 100

Hint: Let x = 20% of student below 8 years.
Let y = student of 8 years age.
Let z = student of above 8 years age.

$$z = \frac{2}{3}y = 48 \Rightarrow y = \frac{3}{2} \times 48 = 72$$

→ 80% of student = $y + z = 72 + 48 = 120$
→ 80% of what number is 120
$80\%k = 120$

$$k = \frac{120}{80\%} = \frac{120}{0.8} = 150$$

Answer: B

12. A student multiplied a number by 3/5 instead of 5/3. What is the approximate percentage error in the calculation?

A. 34% B. 105% C. 75% D. 64%

Hint: Percentage error will be $k = \frac{\frac{5}{3} - \frac{3}{5}}{\frac{5}{3}} \times 100\%$

$$= \frac{25 - 9}{15} \times \frac{3}{5} \times 100\% = \frac{16}{25} \times 100\% = 64\%$$

Answer: D

13. Two students appeared at an examination. one of them secured 9 marks more than the other and his mark was 56% of the sum of their marks. the marks obtained by them are:

A. 39, 30 B. 43, 34 C. 42, 33 D. 41, 32

Hint: Let $x + 9$ and x was their mark.

$(x + 9) + x = 2x + 9$

Since $x + 9$ is 56% of $2x + 9$

$$\Rightarrow x + 9 = \frac{56}{100}(2x + 9) = 0.56(2x + 9)$$

$\Rightarrow x + 9 = 1.12x + 5.04$
$\Rightarrow 1.12x - x = 9 - 5.04 \Rightarrow 0.12x = 3.96$

$$\Rightarrow x = \frac{3.96}{0.12} = 33$$

Thus, $x + 9 = 33 + 9 = 42$

Answer: C

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14. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:

A. 9600 B. 9000 C. 9400 D. 9800

Hint: 9600 is divisible by 15, 25, 40 and 75.

Answer: A

15. What is the missing number in the following series?

1, 9, 25, 49, (…)

A. 54 B. 56 C. 81 D. 64

Hint: $1^2, 3^2, 5^2, 7^2, 9^2$

Answer: C

16. Which of the following is the largest fraction?

A. 7/8 B. 3/4 C. 4/5 D. 7/9

Hint: $\frac{3}{4} < \frac{7}{9} < \frac{4}{5} < \frac{7}{8}$

Answer: A

17. All of the following are ways to write 20 percent of N, **EXCEPT?**

A. 0.20N B. 20N C. 1/5N D. 20/100N

Hint: 20% of $N = \frac{20}{100} \times N = \frac{1}{5}N = 0.2N$

Answer: B

18. The highest among the following is:

A. (0.02) (0.05) C. (0.02) (0.04)
B. (0.02) (0.02) D. (0.02) (0.03)

Hint: $0.02 < 0.03 < 0.04 < 0.05$

Answer: A

19. A circular region has circumference c centimeters and area k square centimeters. If $c = 3k$, what is the radius of the circle in centimeters?

A. 3/2 B. 1/3 C. 2/3 D. 3

Hint: $\frac{\text{circumference}}{\text{Area}} = \frac{c}{k} = \frac{2\pi r}{\pi r^2}$

$\Rightarrow \frac{c}{k} = \frac{3k}{k} = \frac{2}{r} \Rightarrow \frac{r}{2} = \frac{1}{3} \Rightarrow r = \frac{2}{3}$

Answer: C

20. In one month, Hana used 1/6 of her monthly salary for a car payment and ¼ more than the car payment for rent. What fraction of her monthly salary did Hana use that month for the car payment and rent combined?

A. 7/12 B. 3/8 C. 5/12 D. 5/24

Hint: Let x be Han's monthly salary.

$$\Rightarrow \text{fraction} = \frac{(\text{car payment}) + (\text{rent})}{\text{Total}}$$

$$= \frac{\left(\frac{1}{6}\text{of } x\right) + \left(\frac{1}{6}\text{of } x + \frac{1}{4}\text{of } x\right)}{x}$$

$$= \frac{\frac{1}{6}x + \frac{1}{6}x + \frac{1}{4}x}{x} = \frac{\frac{7}{12}x}{x} = \frac{7}{12}$$

Answer: A

21. The population of a town increased from 175,000 to 262,500 in a decade. The average percent increase of population per year is:

A. 5.37% B. 5% C. 7.03% D. 8%

Hint: percent increase $= \frac{262{,}500 - 175{,}000}{175{,}000 \times 10} \times 100\%$

$$= \frac{0.5 \times 100\%}{10} = 5\%$$

Answer: B

22. A fruit seller sold 40% of the oranges he had and still has 420 oranges. How many oranges did he have originally?

A. 700 oranges C. 588 oranges
B. 600 oranges D. 672 oranges

Hint: 60% of $x = 420 \Rightarrow 0.6x = 420$

$$x = \frac{420}{0.6} = 700$$

Answer: A

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Data Interpretation questions (58 – 60)

DIRECTIONS: The following table gives the percentage of marks obtained by seven students in six different subjects in an examination. Answer questions **58 – 60** based on the data in the table and **blacken** the letter of your choice on the separate answer sheet provided.

Student	Maths	Chemistry	Physics	Geography	History	Computer science
Ayele	90	50	90	60	70	80
Aman	100	80	80	40	80	70
Seid	90	60	70	70	90	70
Ruth	80	65	80	80	60	60
Meron	80	65	85	95	50	90
Tekle	70	75	65	85	40	60
Tilahun	65	35	50	77	80	80

23. What is the aggregate of marks obtained by Seid in all the six subjects?

A. 409 B. 419 C. 450 D. 429

Solution: Seid marks = 90 + 60 + 70 + 70 + 90 + 70 = 450

Answer: C

24. In which subject is the overall percentage best?

A. Chemistry B. Maths C. Physics D. History

Answer: B

25. The number of students who obtained 60% and above marks in each subjects is?

A. 2 B. 1 C. 3 D. 4

Hint: Seid and Ruth scored 60% and above in each subject.

Answer: A

SCHOLASTIC APTITUDE (EUEE) 2008

SECTION ONE: ***Verbal Reasoning (Questions 1 – 35)***

This section contains a total of **35** verbal reasoning questions with subsections of analogy, synonyms, antonyms, language usage, sentences completion, reading comprehension and logical reasoning. Each sub-section has its own directions.

Analogy

Definition: A **verbial analogy** expresses a relationship or comparison between sets of words. Normally, analogy contains two pairs of words linked by a word or symbol that stands for an equals (=) sign. A complete analogy compares the two pairs of words and makes a statement about them. It asserts that the relationship between the first pair of words. Is the same as the relationship between the second pair.

DIRECTIONS: For questions **1-6**, identify the relationship that exists between each pair of words presented in capital letters and determine which of the following four pairs of words maintains the same logical relationship. Then **blacken** the letter of your choice on the separate answer sheet provided.

1. **LOGIC: REASONING**

A. ethics : behaviour C. creativity : enthusiasm

B. sensitivity : morality D. bravery : charisma

→ **(for details, see extreme SAT-2009)**

Clue: Use **logic** to **reasoning (logical reasoning)**. Use **ethics** to study **behaviour (goodness /badness)** of something **(ethical behaviour)**.

Answer: A

2. **WEALTH : INCOME**

A. health : medicine C. feeling : affection

B. appetite : hunger D. salary : performance

Clue: If a person owns earns a lot of **income**, he is said to be a **wealthy** person (a rich person) (income results in being wealthy as **performance** results in **salary**).

Answer: D

3. **HEAT : CALORIES**

A. liquid : gallons C. feeling: affection

B. lamp : watts D. salary : performance

Clue: **Heat / energy** is measured by **calories** as **lamp** is measured by **watts**.

Answer: B

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4. **COHABIT : RESIDE**

A. secret : truth C. conspire : plot
A. infiltrate : influence D. coincide : contradict

Clue: These two words **synonym** relationship they mean live together. **conspire** means to secretly **plot** or make plans together.

Answer: C

5. **SAND : GLASS**

A. cotton : wool C. stove : oven
B. seed : flower D. tree : paper

Clue: **Glass** is made up of **sand** whereas **paper** is made up of **tree**.

Answer: D

6. **SEARCH : FIND**

A. sleep : wake C. explore : discover
B. draw : paint D. think : relate

Clue: **Search** and **find** are synonyms as **explore** and **discover** are.

Answer: C

DIRECTIONS: For questions **7-8,** read each of the following words carefully and choose the word that correctly completes the analogy. Then **blacken** the letter of your choice on the separate answer sheet provided.

7. **GUIDE** is **DIRECT** is **REDUCE** is to:

A. increase B. decrease C. preserve D. maintain

Answer: B

8. **SPONGE** is to **POROUS** as **RUBBER** is to:

A. elastic B. solid C. massive D. inflexible

Clue: **sponge** is a soft **permeable** material. (allows fluids /gases to pass through).

AS **rubber** is **inflexible** or **pliable** material.

Answer: D

Synonyms

DIRECTIONS: For questions **9-10,** choose the word that is most nearly **similar** in meaning to the word in capital letters and **blacken** the letter of your choice on the separate answer sheet.

9. **TERMINATE**

A. Depart B. Begin C. End D. Hasten

Synonyms: words of the same /nearly the same meaning.

Clue: **Terminate** is the opposite of **begin. (conclude /end)**

Answer: C

10. **HEARTFELT**

A. Sincere B. Interesting C. Upsetting D. Shocking

Clue: **heartfelt** = felt or believed **deeply** and **sincerely. (genuine, sincere, true)**

Antonyms: are words of opposite meaning.

Answer: A

11. **ALLEVIATE**

A. Transpose B. Fortify C. Exacerbate D. Rectify

Clue: **alleviate** = to make less pain, sever.

exacerbate = to make worse pain, anger (worse or aggravate)

Answer: C

12. **RENAISSANCE**

A. Reawakening B. Regressing C. Revival D. Rebirth

Clue: **renaissance** (A C D) are synonyms.

regressing : going back **(backward)**

Answer: B

Note: (See Extreme SAT-2009)

Language Usage

DIRECTIONS: For questions **13-14,** below choose the choice that most nearly has the same meaning as the **underlined** word from the given alternatives and **blacken** the letter of your choice on the separate answer sheet provided.

13. The technical department enthusiastically hired W/ro Abebech because she was **proficient** in the use of computers.

A. sincere B. skilled C. smart D. adequate

Clue: **proficient:** skilled, fluent, practiced

Answer: B

14. The assistant was fast becoming an **indispensable** member of the department, sot hey had no choice but to offer him a higher salary to stay on.

A. essential B. active C. determined D. creative

Clue: **indispensable:** essential, very important.

Answer: A

Sentence completion

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DIRECTIONS: Each sentence (questions **15-20**) below has two blanks. Each blank indicates that something has been omitted. Beneath the sentence are four sets of words. Choose a word or the set of words that best fits the meaning of the sentence as a whole and **blacken** the letter of your choice on the separate answer sheet provided.

15. She based her conclusion on what she _____ from the _____, not on what she imagined.

A. Inferred : rumor
B. assumed : hearsay
C. understood : gossip
D. inferred : evidence

Answer: D

16. Although the formation of a union was not _____ by the management of the organization, it was _____ that any effort on the part of employees to unionize would not be welcome.

A. prevented : imperceptible
B. banned : implicit
C. facilitated : infamous
D. commandeered : calculated

Answer: B

17. Although opponents of the expansion of the market economy are in _____, they continued to constitute _____ political force through the century.

A. retreat : a powerful
B. error : an inconsequential
C. disarray : a disciplined
D. jeopardy : an ineffective

Answer: A

18. Man is a _____ animal, and much more so in his mind than in his body: he may like to go alone for a walk, but he hates to stand along in his _____.

A. gregarious : opinions
B. solitary : thoughts
C. conceited : vanity
D. singular : uniqueness

Answer: A

19. You cannot become _____ teacher without completing the _____ student-teaching assignment.

A. certified : prerequisite
B. preferred : optional
C. required : certified
D. prerequisite : advisable

Answer: A

20. The teacher is becoming _____ with the unacceptable _____ of some students in class.

A. patient : gifts
B. impatient : behavior
C. happy : size
D. indifferent : gifts

Answer: B

Note: For better understanding about **sentence completion questions**, see **Extreme SAT-2009**

DIRECTION: In questions **21** and **22**, an incomplete statement followed by fillers is given. Choose the filler which completes incomplete stem correctly and meaningfully. Then, **blacken** the letter of your choice on the separate answer sheet provided.

21. The weather outside was extremely pleasant and hence we decided to...

A. enjoy a morning ride in the open.
B. utilize our time in watching the television.
C. refrain from going out for a morning walk.
D. employ this rare opportunity for writing letters.

Clue: The fine weather is very pleasing to enjoy riding in the morning.

Answer: A

22. Despite his best efforts to conceal his anger,....

A. he failed to give us an impression of his agony.
B. we would detect that he was very happy.
C. people came to know that he was annoyed.
D. he succeeded in camouflaging his emotions.

Clue: It is about the opposite /contrast ideas between two sentences.

➤ **agony (anguish, torment, suffering.**

Answer: A

DIRECTIONS: Read Passages I, II and III below and answer the questions that follow each passage on the basis of what is stated or implied in the passages. Then, **blacken** the letter of your choice on the separate answer sheet provided.

Reading Passage I

The Earth's past climate—including temperature and elements in the atmosphere—has recently been studied by analyzing ice samples from Greenland and Antarctica. The air bubbles in the ice have shown that, over the past 160,000 years, there has been a close correlation between temperature changes and level of natural greenhouse gases carbon dioxide and methane. One recent analysis from Greenland showed that at the end of the last glacial period (when the great ice sheets began to retreat to their present position), temperatures in southern Greenland rose from 5 to 7 degrees in about 100 years. Air bubbles are not the only method of determining characteristics of the Earth's ancient climate history. Analysis of dust layers from ancient volcanic

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activity is another such method; as is the study of ice core, which interpret past solar activity that may have affected out climate.

23. Which of the following is true about Greenhouse gases, according to the passage?

A. Their level correlates with changes in temperature.
B. They include carbon dioxide and methane only.
C. Their analysis has little or no meaning in explaining change in the earth's climate.
D. there is nothing to worry about their effect on the earth's climate.

Answer: A

24. Which of the following is true according to the passage?

A. Change in the earth's climate is a phenomenon that relates to changes in the atmosphere.
B. The great ice sheets began to retreat to their present position at the end of the last glacial period.
C. There is no clear association between the atmospheric changes and climate of the earth.
D. Our climate may be influenced by the past polar activates.

Answer: A

Reading Passage II

Many people who have come close to death from drowning, cardiac arrest or other causes have described near-death experiences—profound, subjective events that sometimes result in dramatic changes in values, beliefs, behavior, and attitudes toward life and death. these experiences often include a new clarity of thinking, a feeling of wellbeing, a sense of being out of the body, and visions of bright light or mystical encounters. Such experiences have been reported by an estimated 30 to 40 percent of hospital patients who were revived after coming close to death and about 5 percent of adult Americans in a nationwide poll. Near-death experiences have been explained as a response to a perceived threat of death (a psychological theory); as a result of biological states that accompany the process of dying (a physiological theory); as a result of biological states that accompany the process of dying (a physiological theory); and as a foretaste of an actual state of bliss after death (a transcendental theory).

25. The phrase *dramatic change* in the passage refers to:

A. passionate changes
B. theatrical changes
C. profound changes
D. exaggerated changes

Answer: D

26. The primary purpose of this passage is to:

A. entertain patients
B. inform readers
C. persuade relatives
D. express disbelief in the afterlife

Answer: D

Reading Passage III

Lions are rather reticent about expanding their energy in hunting. More than three quarters of kills are made by lionesses holding front position and carefully scanning ahead. The cubs lag playfully behind and the males bring up the rear and walk slowly with their massive heads nodding with each step as if they were bored with the whole matter. With lionesses' busy hunting, the males function as guard for the cubs, protecting them particularly from hyenas.

27. Male Lions protect their cubs

A. from the members of their own species.
B. from hyenas only.
C. more from hyenas than from other animals.
D. from hyenas as much as from other enemies.

Answer: C

28. Lioness go for hunting

A. all alone.
B. with their cubs only.
C. with their cubs and male partners.
D. with their male partners only.

Answer: A

29. When the lionesses go in search for their prey, they are very

A. serious B. sluggish C. playful D. cautious

Answer: C

30. According to the passage male lions generally do not go for hunting because they:

A. wish to save their vigor for other things.
B. they are very lazy.
C. do not like it.
D. want lioness to get training.

Answer: A

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Logical reasoning

In each question below is given a statement followed by two conclusions numbered I and II. You have to assume the statement true, then consider the conclusions and decide which of them logically follows beyond a reasonable doubt from the information given in the statement. Then, **blacken** the letter of your choice on the separate answer sheet provided.

31. **Statement:** Population increase coupled with depleting resources is going to be the scenario of many developing counties in days to come.

Conclusion: I. The population of developing countries will not continue to increase in future.

II. It will be very difficult for the governments of developing countries to provide its people decent quality of life.

A. Only conclusion I follows
B. Only conclusion II follows
C. Both I and II follow
D. Neither I nor II follows

Answer: B

32. **Statement:** Never before such a lucid book was available on the topic.

Conclusion: I. Some other books were available on this topic.

II. You can write lucid books on very few topics.

A. Only conclusion I follows
B. Only conclusion II follows
C. Both I and II follow
D. Neither I nor II follows

Answer; B

33. **Statement:** Be humble even after being victorious.

Conclusion: I. Many people are humble after being victorious.

II. Generally people are not humble.

A. Only conclusion I follows
B. Only conclusion II follows
C. Neither I nor II follows
D. Both I and II follow

Answer: B

34. **Statement:** Apart from the entertainment value of television, its educational value cannot be ignored.

Conclusion: I. People take television to be a means of entertainment only.

II. The educational value of television is not realized properly.

A. Only conclusion I follows
B. Only conclusion II follows
C. Neither I nor II follows
D. Both I and II follow

Answer: D

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Directions: Questions 1 – 5 deal with analogy. In question 1, a related pair of words written in capital letters is followed by four pairs of words or phase. Select the pair that best expresses a relationship similar to that expressed in the original capitalized pair. In questions 2 – 5 related pair of words are followed by a word that has similar relationship when paired with another word. Answer these questions by selecting the choice that complete the given analogy correctly by shading the letter of the correct choice on the separate answer sheet.

1. OPINION is to DEBATE as HYPOTHESIS is to _____.
   A. Rejecting C. Stating
   B. Discovering D. Testing
   Clue: People forward their opinion during a debate somebody works to test the hypothesis in different situations to explain (for example why foods are more likely to cause problems).
   Answer: D

2. LAND is ARABLE as river is to _____.
   A. Shallow C. Water
   B. Navigable D. Impenetrable
   Clue: Arable land is land that is used for arable farming good for only crops growing rather than fruits. Navigate river is made and deep enough for aboat to travel along safely. **Answer: B**

3. DEPARTURE: ARRIVAL
   A. Inception: Conclusion C. Upshot: End
   B. Beginning: Culmination D. Process: Outcome
   Clue: The two words have opposite meaning (they are anonyms).
   - **Departure**: the act of somebody's leaving a job or a member of organization.
   - **Arrival**: the act of somebody's starting a new job.

   Beginning: starting = culmination: end/termination.
   Answer: B

4. WIDOW is to a WOMAN as _____ is to a MAN.
   A. Divorce C. Widow
   B. Widower D. Finance
   Clue: Widow is a woman whose husband has died. Fiancé is a man to whom a woman is engaged to be married.
   Answer: D

5. CAUSE to EFFECT as PREDECESSOR is to _____.
   A. Foundation C. Successor
   B. Outcome D. Precursor

Clue:
- **Predecessor:** a person who had your job **before** you.
- **Successor:** a person who had your job after you.
- **(Sequence relationship)**

Answer: C

Directions: Questions 6 – 7 consist of a word printed in capital letters, followed by four words. Choose the word that is most nearly opposite in meaning to the word in capital letters and blacken the letter of your choice on the separate answer sheet.

6. ARROGANCE
   A. Truthful C. Courage
   B. Humility D. Self – esteem
   Clue: Arrogance (N): an overbearing manner
   Humility (N): not proced; modesty
   Answer: B

7. THRIVE
   A. Trail C. Deteriorate
   B. Strive D. Distractive
   Clue: Thrive (v): to grow, to increase, to flourish, to prosper (be profitable)
   Deteriorate (v): to make worse, to impair, to degenerate.
   Answer: C

SYNONYMS

Directions: Questions 8 – 9 consist fo words printed in capital letters followed by four words. Choose the word that is most nearly similar in meaning to the word in capital letters. Then, find the question number and blacken the letter of your choice on the separate answer sheet.

8. IMMEDIATELY:
   A. Instantly C. Slowly
   B. Gradually D. Temporarily
   Clue: Immediately (adv): Instantly or without delay.
   Answer: A

9. CURTAIL
   A. Facilitate C. Recover
   B. Improve D. Restrain
   Clue: curtail (v): control, limit, restrain
   Answer: D

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LANGUAGE USAGE

Directions: Each of the Questions below (10 – 12) consists of underlined phrase or word. From the given alternatives, choose the one that most nearly has the same meaning as the underlined phrase or word and blacken the letter of your choice on the separateanswer sheet.

10. Taciturn by nature, Eyasu speaks very little even to his own family members.
    A. Quiet C. Warm
    B. Talkative D. Obtrusive
    Clue: Tacitrun (adj): silent, disinclined to speak, reticent, untalkative. (quiet) **Answer: A**

11. If you can't make it to my graduation party, it's not a big deal.
    A. Not tolerable C. Surprising
    B. No need to worry D. Something I dislike
    Clue: If you say "**big deal**" to someone to show that you are not impressed by something that they consider important or impressive. **Answer: C**

12. Hannan remained as cool as a cucumber even when she was in the pressure of the competition against her opponent.
    A. Strong C. Calm
    B. Ruffled D. Agitated
    Clue: As cool as cucumber (idiom): very relaxed, calm and no emotional. **Answer: C**

Sentence Completion

Directions: Each sentence (13 – 17) below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are four words or set of words. Choose the word or set of words for each blank that best fits the meaning of the sentence as a whole and blacked the letter of your choice on the separate answer sheet.

13. As Mohammed and Abdi have planned to join university in the coming year by sponsoring themselves, they are _____ and _____ a lot.
    A. Reading – saving C. Teaching – Learning
    B. Applying – paying D. Working – entertaining
    Clue: The university's requirements are very stiff. Therefore, the should be ready both mentally and financially by reading and saving a lot. **Answer: A**

14. The _____ of the apartment was unbelievable; it was difficult to realize that human beings could live in such _____.
    A. Squalor – filth C. Disorder – Isolation
    B. Stench – disarray D. Spacious – Proximity
    Clue: Squalor: a very dirty, an unpleasant condition
    Filth: a disgusting amount of dirt **Answer: A**

15. Even after long years of extensive investigation, scientists could not discover the treatment for cancer. The treatment still remains _____.
    A. Clear C. Apparent
    B. Cursory D. Mysterious
    Clue: Mysterious (adj): of unknown origin. **Answer: D**

16. While the disease is in _____ state it is almost impossible to determine its existence by _____.
    A. A critical – examination C. A latent – observation
    B. An acute – analysis D. A dormant – postulate
    Clue: While the disease is in a dormant state.
    - **Dormant (adj):** inactive, asleep, suspended its most impossible to determine its existence by **postulate.**
    - **Postulate:** to assume without proof.
    Answer: D

17. Because artist Aster Aweke paved the way for later jazz-fusion musicians, she is considered as _____ of the style.
    A. Beneficiary C. Revivalist
    B. Connoisseur D. Progenitor
    Clue: Connoisseur (N): Is someone who knows a lot about the arts, food, drink or some other subjects.
    - A specialist of a given field whose opinion is valued, especially in one of the fine arts. **Answer: B**

Direction: Read passages I and II below carefully and answer questions 18 – 25 based on the information stated or implied in the passages. Then find the question number and blacken the letter of your choice on the separate answer sheet.

Reading Passage I

Black holes are the most efficient engines of destruction known to humanity. Their intense gravity is a one – way ticket to oblivion, and material spiraling into them can heat up to millions of degrees and glow

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brightly. Yet, they are not all – powerful. Even super-massive black holes are minuscule by cosmic standards. They typically account for less than one percent of their galaxy's mass. Accordingly, astronomers long assumed that super – massive holes, let along their smaller cousins, would have little effect beyond their immediate neighborhoods. So it has come as a surprise over the past decade that black hole activity is closely intertwined with star formation occurring farther out in the galaxy.

18. Which of the following most resemble the relationship between "black hole activity" and "star formation" (lines 8 – 9) as described in the passage?
   A. At drought in a wildness area causes a significant loss of vegetation in that area.
   B. Overfishing in a gulf leads to an increase in the population of smaller aquatic organisms.
   C. A volcanic eruption on one continent results in higher rainfall totals on another continent.
   D. Industrial emissions in one region lead to an increase in airborne pollutants in adjacent regions.

Answer: D

19. What does the word "minuscule" in line 4 refer to?
   A. Tiny
   B. Very large
   C. Invisible
   D. noticeable

Clue: Minuscule (adj): meaning very smell (tiny).

Answer: A

20. What is the primary purpose of the passage?
   A. Advocating the importance of black holes.
   B. Disproving an existing theory of black holes.
   C. Verifying the danger of black holes for human beings.
   D. Explaining the role that black holes play in the formation of stars.

Answer: D

21. Which of the following best describes the function of the statements in lines 7 – 9 ("So it . . . galaxy")
   A. It challenges recent scientific findings.
   B. It offers examples to support a theory.
   C. It introduces a new view of information presented earlier in the passage.
   D. It provides support for the argument asserted in the preceding statement.

Answer: D

Reading Passage II

Born in Adigrat, Tigray Region of Ethiopia, MirutsYifter spent early parts of his youth working in different factories as a carriage driver. His talent as a long – distance runner was noticed when he joined the Ethiopian Air Force.

Miruts was called to the Ethiopian National Team for the 1968 Summer Olympics in Mexico City, but he made his Olympic **debut** four years later in Munich Olympics where he won a bronze medal in 10,000 meters. In the 1973 All – Africa Games he won one gold and one silver medals in 10,000 and 5,000 meters, **respectively**. A the first African Championships in 1979 he won two gold medals in 5,000 and 10,000 meters.

Miruts was unable to participate in the 1976 Summer Olympics because his nation boycotted the even. Four years later in Moscow, Mirutsmade **up** of this disappointments. In the final of the 10,000 meters he **sprinted** into the lead 300 meters from the finish and won by ten meters difference from his follower. Five days later, in the 5000 meters final, Miruts was **boxed in** during the last lap. But with 300 meters to go, his Ethiopian teammate, Mohammed Kedir, **stepped aside** and Miruts again sprinted to **victory**. Due to his **abrupt** change in speed when **executing** his kick to the finish, Miruts acquired the nickname "Miruts the shifter."

MirutsYifter died at the age of 72 on 22 December 2016 in Toronto, Canada where he had lived since 2000 after suffering from **respiratory** problems.

22. According to the passage, which of the following is TRUE about MirutsYifter?
   A. Miruts had won two gold medals in one Olympic.
   B. Miruts had run in all Olympics since Munich Olympics.
   C. Miruts won a gold medal the first time he ran at the international stage.
   D. Mirtus began his career as long-distance runner since his childhood years.

Answer: A

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23. Which of the following is closest in meaning to the word abrupt (line 14)?

A. Easy
B. Difficult
C. Strong
D. Sudden

Clue: An **abrupt** change or event is a very **sudden**.

Answer: A

24. According to the passage, which one of the following is TRUE about the gold medal in 5,000 meters Miruts won in Moscow Olympics?

A. Miruts had to run with knee injury when he won the gold medal.
B. It would not have made any difference whether or not Mohammed Kedir had stepped aside.
C. Miruts was in uncomfortable position but he came out of that position without anyone's help.
D. Thanks to the support from Mohammed Kedir, Miruts won the medal with his unique sprint at the end.

Answer: D

25. Which of the following is closest in meaning to the word debut (line 5)?

A. First appearance
B. Second appearance
C. Usual appearance
D. Obligatory appearance

Clue: **debut (N):** A performer's first performance to the public.

Answer: A

Local and Analytical Reasoning

Directions: Questions 26 – 35 deal with logical and analytical reasoning. Read the given set of conditions and assumptions for each question. Choose the answer that you think is most appropriate among the given alternatives.

26. If you present a yellow pass, then you may enter the compound. If the statement above is true, which of the following must also be true?

I. If you do not present a yellow pass, then you may not enter the compound.
II. If you may enter the compound, then you must have presented a yellow pass.

III. If you may not enter the compound, you did not present a yellow pass

A. III only
B. II only
C. I only
D. I and II only

Clue: The opposite is also true. **Answer: C**

Questions 27 – 28 are based on the following paragraph.

A company's facilities were too small to accommodate the crowd for a special one day meeting. The company members received a license to use the public park, and a public address system was promptly installed. Citizens who were not members of that company protested the action of the park department for having issued this license.

27. Which of the following is the best argument used by the citizens who protested the action by the park department?

A. Freedom of speech was violated.
B. A public referendum should have been held.
C. Citizens would be denied the enjoyment of the public park during this one day.
D. The town would incur expenses to clean up the park after the one day meeting.

Answer: B

28. What is the best argument used by the company to retain its license?

A. The company is a local taxpayer.
B. No one would be excluded from entering the park during the meaning.
C. Wide publicity had already been given to the location of special meeting.
D. The park had been used before by companies for meetings, concerts, rallies.

Answer: A

29. Since it is possible that substances contained in certain tree roots may provide a cure of caner, the government must provide sufficient funds to allow through testing of this possibility.

A. A cure for cancer would be extremely valuable society.
B. Substances contained in certain tree roots will probably cure cancer.
C. The possibility of finding a cure is sufficient reason for funding research into possible cancer cures.
D. The line of research mentioned offers at present the most promising possibility for finding a cure for cancer.

Answer: C

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Questions 30 – 31 are based on the following conditions.

Ms. Halima: Mr. Abera insists that the only way for our company to increase its profits is to double he advertising budget. That obviously is not the answer. Our two major competitors have operations similar to ours. Both are showing increased profits while spending less on advertising than we presently spend.

30. Which of the following statements would be Mr. Abera's most effective rebuttal of Ms. Halima's argument:
    A. My proposal is not 'obviously' wrong. There is only one way to find out if it is wrong, and that is to try it.
    B. You have offered statistics to back up neither your claims nor nay proposal for an alternatives solution to our problem.
    C. I have been in this business for 30 years, during which time I have repeatedly proven my ability to identify and solve business problems.
    D. Our two major competitors do not need to advertise as much as we do, because they are already much better known and have large shares of the market.

Answer: A

31. Ms. Halima's primary method of making her point is to:
    A. Draw an analogy.
    B. Question Mr. Abera's competence.
    C. Present evidence which was previously overlooked.
    D. Suggest a different underlying cause of the problem.

Answer: A

Questions 32 – 35 are based on the following information

In order to conduct the work of a mail order concern it is necessary to have a minimum of three workers each day. The staff consists of five persons who work on a part – time basis. Mohammed can work on Mondays, Wednesdays. Nati can report for work on Meaza cannot report for work on Wednesdays. Nati can report fro work on Tuesdays and Wednesdays only. Almayehu cannot work on Fridays. Yeshi is available anytime except on the first Monday and Tuesday of the month.

32. During which day of the week might it be impossible to obtain a full complement of workers?
    A. Monday C. Thursday
    B. Tuesday D. Wednesday

Answer: C

33. Which three are available on any Monday?
    A. Meaza, Nati and yeshi
    B. Mohammed, Yeshi, and Nati
    C. Nati, Meaza, and Mohammed
    D. Alemayehu, Meaza, and Mohammed

Answer: D

34. During which day of the week would it be necessary to call on Mohammed to complete the full complement of workers?
    A. Friday C. Thursday
    B. Monday D. Wednesday

Answer: C

35. Which three could you count on to report for work on Friday?
    A. Meaza, Nati, and Yeshi.
    B. Nati, Meaza, and Mohammed.
    C. Mohammed, Meaza, and Yeshi.
    D. Mohammed, Meaza, and Alemayhu

Answer: C

Section two: Quantitative Reasoning (Questions 36 – 60)

Directions: Solve each of the problems given in questions 36 – 54 below and select the best answer from the given alternatives. Then find the correct question number and blacken the letter of your choice on the separate answer sheet provided.

36. When a number is divided by 2 the result is equal to the result when that same number is divided by 4. What is that number?
    A. -2 B. 0 C. 2 D. 4

Solution: Let x be the number and k – be the result

$$\frac{x}{2} = k \ \ldots * \text{ and } \frac{x}{4} = k \ \ldots **$$

From * and **

$$\frac{x}{2} = \frac{x}{4} \Rightarrow 4x = 2x$$

$$\Rightarrow 4x - 2x = 0 \Rightarrow x = 0$$

Answer: B

37. If $\frac{6}{x} = \frac{18}{x+6}$, what is the value of $\frac{x}{3}$?
    A i B. 2 C. 3 D. 4

Solution: $\frac{6}{x} = \frac{18}{x+6} \Rightarrow 6(x+6) = 18x$

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$\Rightarrow 6x + 36 = 18x \Rightarrow -12x = -36$ then $x = 3$

If $x = 3$ then $\frac{x}{3} = \frac{3}{3} = 1$ **Answer: A**

38. What is the value of $x + 10$, if $(x - 5) = \frac{11}{(x+5)}$, for $x \neq -5$?

A. -10 B. 4 C. 10 D. 25

Solution: $x - 5 = \frac{11}{x+5} \Rightarrow (x-5)(x+5) = 11$

$\Rightarrow x^2 - 25 = 11 \Rightarrow x = \pm\sqrt{36} = \pm 6$

i) If $x = -6$ then $x + 10 = -6 + 10 = 4$

ii) If $x = 6$ then $x + 10 = 6 + 10 = 16$

Answer: B

39. Which of the following is less than z if z is a number between 0 and 1?

A. z^2 B. $\sqrt{z}$ C. $2z$ D. $1/z$

Hint: Let $z = \frac{1}{4}$ then $z^2 < z < \sqrt{z}$ **Answer: A**

40. What is the sum of the following values
(1/4 + 0.25 + 25/100 + 10/40 + 25/50 + 50/100)?

A. 1 B. 2 C. 3 D. 4

Solution: $\frac{1}{4} + 0.25 + \frac{25}{100} + \frac{10}{40} + \frac{25}{50} + \frac{50}{100}$

$= \frac{1}{4} + \frac{25}{100} + \frac{25}{100} + \frac{10}{40} + \frac{25}{50} + \frac{50}{100}$

$= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{2} + \frac{1}{2} = \frac{4}{4} + \frac{2}{2} = 1 + 1 = 2$

Answer: B

41. What is the value of $(0.0001 \times 20^4) \div 16/10$?

A. 1 B. 4 C. 10 D. 100

Hint: $\left(10^{-4} \times 2^4 \times 10^{-4}\right) \times \frac{10}{16} = 10$

Answer: C

42. If 75 percent of x is equal to p percent of 25, where p > 0, what is the value of x/p?

A. 1/3 B. 3/16 C. 3/4 D. 3

Hint:

75% of x = P % of 25 $\Rightarrow \left(\frac{75}{100}\right)x = \left(\frac{P}{100}\right)25$

$\Rightarrow 75x = 25P \Rightarrow \frac{x}{P} = \frac{25}{75} = \frac{1}{3}$ **Answer: A**

43. If $15 + x$ is 5 more than 20, what is the value of x?

A. 5 B. 10 C. 25 D. 50

Solution: $15 + x = 20 + 5 \Rightarrow 15 + x = 25$ then $x = 25 - 15 = 10$

Answer: B

44. When 80,000 is written as 8.0×10^n, what is the value of n?

A. 2 B. 3 C. 4 D. 5

Solution: $80{,}000 = 8.0 \times 10^4$ then $8.0 \times 10^n = 8.0 \times 10^4$

$\Rightarrow 10^n = 10^4 \Rightarrow n = 4$

Answer: C

45. The price for one quintal of sugar is birr 5,800 and it decreases at the rate of birr 2 every three months. The price for one quintal of coffee is birr 4,200 and it increases at the rate of birr 8 every four months. How many years will it take the two commodities to become equal in price?

A. 10 B. 20 C. 50 D. 60

Solution: The price of one quintal sugar is birr 5800 and decreases at rate of birr 2 every 3 months 5,800, 5792, ... formed an arithmetic sequence $d = -8$.

$A_8 = 5800 + (n-1)(-8) = 5808 - 8n$ ... *

The price of coffee of one quintal is 4200, 4176, ... forms arithmetic sequence $d = 24$, $A_n = 4200 + (n-1)\,24 = 4176 + 24n$

$32n = 1632 \Rightarrow n = 51$ yrs

$t = n - 1 = 50$ yrs **Answer: C**

46. If as many 7 – meter pieces of wire as possible are cut from a wire that is 36 meters long, what is the total length of the wire that is left over?

A. 1 meter C. 3 meters

B. 2 meters D. 4 meters

Hint: $7x = 36 \Rightarrow x = 36 \div 7 = 5$, Reminder 1 **Answer: A**

47. Hawa used ¼ of her monthly salary for buying teff and ½ more than the price of teff for house rent. What fraction of her monthly salary did Hawa use that month for the teff and rent combined?

A. 1/3 B. 7/12 C. 5/8 D. ¾

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Solution: Let x – be Hawa's monthly salary

Hawa spend for teff $= \frac{1}{4}x$

Hawa spend for house rent $= \frac{1}{4}x + \frac{1}{2}\left(\frac{1}{4}x\right)$

Hawa spend for teff and house rent $= \frac{\frac{x}{4}+\frac{x}{4}+\frac{x}{8}}{x}$

Hawa's combined spend $= \frac{\frac{2x+2x+x}{8}}{x} = \frac{5x}{8x} = \frac{5}{8}$ **Answer: C**

48. When 5 times the number x is added to 12, the result is 32. What number results when 2 times x is added to 7?

A. -15 B. -5 C. 5 D. 15

Solution: $5x + 12 = 32$ then $2x + 7 =$ _____

$\Rightarrow 5x = 32 - 12 = 20$

$\Rightarrow 5x = 20$ then $x = 4$; since $x = 4$, then $2x + 7 = 2(4) + 7 = 15$

Answer: D

49. If three students have scores X, Y and 80, with their sum equal to 180, what is the average (arithmetic mean) of the three scores?

A. 60 B. 70 C. 80 D. 90

Solution: The sum of three scores is $x + y + 80 = 180$ then the

$$\text{average} = \frac{x+y+80}{3} = \frac{180}{3} = 60$$

Hence arithmetic mean is 60. **Answer: A**

50. If the sum of three consecutive integers is 33, what is the middle integer?

A. 9 B. 10 C. 11 D. 12

Solution: Let n – be any integer number then n, n + 1, n + 2 are three consecutive integers $\Rightarrow n + n + 1 + n + 2 = 33$

$\Rightarrow 3n + 3 = 33 \Rightarrow 3n = 30$ then $n = 10$, $n + 1 = 11$ and $n + 2 = 12$

Hence the three consecutive integers are (n, n + 1, n + 2) = (10, 11, 12) the middle integer is 11.

Answer: C

51. What is the value of E, if ABCDE is a five digit positive number divisible by 5 and 10?

A. 0 B. 1 C. 2 D. 5

Solution: ABCDE is divisible by 5 and 10 if and only if the unit digit E = 0. **Answer: A**

52. The arithmetic mean of p and q is 15, and the average of y and x is 15. What is the average of p, y, x and q?

A. 7.5 B. 15 C. 22.5 D. 30

Solution: The arithmetic mean of p and q is:

$$\frac{p+q}{2} = 15 \Rightarrow p + q = 30 \ldots *$$

The arithmetic mean of x and y is :

$$\frac{x+y}{2} = 15 \Rightarrow x + y = 30 \ldots **$$

From * and ** the arithmetic mean of p, q, x and y is

$$\frac{p+q+x+y}{4} = \frac{30+30}{4} = \frac{60}{4} = 15$$ **Answer: B**

53. Which of the following is the average of the first 10 odd numbers?

A. 10 B. 15 C. 20 D. 25

Solution: The first 10 odd numbers are 1, 3, 5, ..., 19 forms an arithmetic mean with $A_1 = 1$, $A_{10} = 19$ and the common difference d = 2 then the sum of the numbers

$$S = \frac{n}{2}(A_1 + A_n) = \frac{10}{2}(1+19) = 100$$

Then the average $= \frac{5}{10}(20) = \frac{100}{10} = 10$ **Answer: A**

54. A number is called "even – odd" if it is halfway between an even integer and an odd integer. If x is an even – odd number, which of the following must be true?

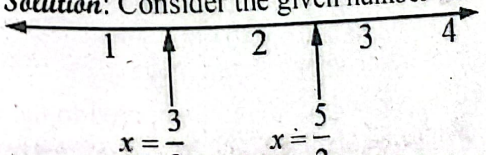
I. $2x$ is an integer. II. $2x$ is even – odd

III. x is halfway between two even integers

A. I only C. I and II only

B. II only D. II and III only

Solution: Consider the given number line



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$$\left.\begin{aligned} 2x &= 2\left(\frac{3}{2}\right) = 3 \\ 2x &= 2\left(\frac{5}{2}\right) = 5 \end{aligned}\right\} \text{ is an integer}$$

Answer: A

Directions: Questions 55 – 60 below are based on the hypothetical data given in the table below.

Questions 55 – 60 are based on the hypothetical data given below.

Annual Income and Expenditure of Five Schools

School	Income (in ETB)	Expenditure (in ETB)
A	1,200,000	1,000,000
B	1,400,000	1,200,000
C	1,000,000	1,100,000
D	2,000,000	1,600,000
E	1,600,000	1,100,000
Sum	**7,200,000**	**6,000,000**

55. How many schools had higher rate of saving compared to the rate of the saving from the total income of all school?

A. 1 B. 2 C. 3 D. 4

Solution: Rate of saving school A = each school income – each school expenditure

Rate of saving A = 1,200,000 – 1,000,000 = 200,000

Rate of saving school B = 1,400,000 – 1,200,000 = 200,000

Rate of School C = 1,000,000 – 1,100,000 = –100,000

Rate of saving school D = 2,000,000 – 1,600,000 = 400,000

Rate of saving school E = 1,600,000 – 1,100,000 = 500,000

Total rate of saving = total income – total expenditure

= 7,200,000 – 6,000,000 = 1,200,000 then

the rate of saving $= \frac{1,200,000}{5} = 240,000$

∴ rate of saving at A = 200,000 < 240,000

Rate of saving at B = 200,000 < 240,000

Rate of saving at C = 100,000 < 240,000

Rate of saving D = 400,000 > 240,000

Rate of saving at E = 500,000 > 240,000

School D and E had higher rate of saving compared to the rate of saving from the total income of all school. **Answer: B**

56. If each school was expected to save at least 10% of its income, how many schools have met the expectation?

A. 1 B. 2 C. 3 D. 4

Solution: School A = 1,200,000 × 10% = 120,000 birr

School B = 1,400,000 × 10% = 140,000 birr

School C = 1,000,000 × 10% = 100,000 birr

School D = 2,000,000 × 10% = 200,000 birr

School E = 1,600,000 × 10% = 160,000 birr

So, school D and E met the expectation because school D: 400,000 > 200,000 and school E: 500,000 > 160,000.

Answer: B

57. Assume that one school will be awarded based on the proportion of saving to the income generated. Which school do you think deserves the award?

A. B B. C D. D D. E

Solution:

From the above two question school C saving to income generated proportion i.e., saving = 1,000,000 × 10% = 100,000 and income generated = 1,000,000 – 1,100,000 = –1,000,000

Answer: B

58. Which one of the following is **NOT** TRUE about the income – expenditure data given above?

A. The difference between income and expenditure was the highest for School E.

B. All schools have managed to keep their expenditure lower than their income.

C. Most of the schools had annual income which was larger than the average annual income of the five schools combined.

D. Only one school had an annual expenditure which was larger than the average annual expenditure of the five schools combined.

Solution:

A. The difference between income and expenditure = 1,000,000 – 1,100,000 = 500,000 was the highest for E than the rest of the schools. **True**

B. All school have managed to keep their expenditure lower than their income. **True**

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C. Most of the schools had annual income which was larger than the average annual income of the five schools combined.

The average annual income of the five schools:

$= \frac{7,200,000}{5} = 1,440,000$

Therefore school A annual income < the average annual income i.e., 1,200,000 < 1,440,000

School B annual income < the average annual income i.e., 1,400,000 < 1,440,000

School C annual income < the average annual income i.e., 1,000,000 < 1,440,000

School D annual income < the average annual income i.e., 2,000,000 > 1,440,000

School E annual income > the average annual income

∴ Most of the schools had annual income which was larger than the average annual income of the five school is FALSE.

D. Only one school had an annual expenditure which was larger than the average annual expenditure of the five schools combined. TRUE because average annual expenditure of five schools.

$= \frac{6,000,000}{5} = 1,200,000$

Annual expenditure school D > the average of all the school expenditure i.e., 1,600,000 > 1,200,000

Answer: C

59. The difference between the average annual income and the average annual expenditure of the five schools in ETB is _____.

A. 100,000 C. 200,000

B. 240,000 D. 1,200,000

Solution: Average annual income $= \frac{7,200,000}{5} = 1,440,000$

Average annual expenditure $= \frac{6,000,000}{5} = 1,200,000$

Hence the difference between average annual income and the average annual expenditure is:

= 1,440,000 – 1,200,000 = 240,000 **Answer: B**

60. The average annual income of the five schools in ETB is _____.

A. 1,200,000 C. 1,300,000

B. 1,400,000 D. 1,440,000

Solution: Average annual income $= \frac{7,200,000}{5} = 1,440,000$

Answer: D

61. **Tip**: The sum of the ages of 5 children born at the interval of 3 years each is 50 years what is the age of the youngest Child?

A. 4 years B. 8 years C. 10 years D. 12 years

Solution: Le the ages of the children be y,

$(y + 3), (y + 6) + (y + 9)$ and $(y + 12)$ years then

$y + (y + 3) + (y + 6) + (y + 9) + (y + 12) = 50$

$5y + 30 = 50 \Rightarrow 5y = 20 \Rightarrow y = 4$

∴ Age of the youngest child is 4yrs. **Answer: A**

62. **Tip**: A boy was asked to multiply a number by 25. He instead multiplied the next number by 52 and got the answer 324 more than the correct answer the number to be multiplied was:

A. 10 B. 12 C. 15 D. 25

Solution: Le the number be x then $52x - 25x = 324$ i.e., $27x = 324$

$\Rightarrow x = 12$ **Answer: B**

63. **Tip**: If the sum of two numbers is 22 and the sum of their squares is 404, then the product of the numbers is

A. 40 B. 44 C. 80 D. 88

Solution: Let the number be x and y then, $x + y = 22$ and $x^2 + y^2 = 404 = 80$

$\Rightarrow 2xy = (x + y)^2 = (x^2 + y^2) = 22^2 - (404) = 484 - 404$

$\Rightarrow xy = \frac{80}{2} = 40$ **Answer: A**

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DIRECTIONS: For questions 1 – 7 red each of the following words carefully and choose the word that correctly completes the analogy. Then, **blacken** the letter of your choice on the separate answer sheet provided.

1. ADULT is to CHILD as CAT is to:
   A. KITTEN B. MARE C. HERD D. ANIMAL
   Clues: A **child** grows to an **adult** (a fully grown person).
   And a **kitten** (young cat) grows to a **cat**, a fully grown cat. (**Age** relationship) **Answer: A**

2. CAP is to HEAD as RING is to:
   A. EAR B. LEG C. HAND D. NECK
   Clues: A **cap** is a soft hat which is worn on the **head** by men and boys. A **ring** is a piece of jewelry that is worn on the **hand** (wearing and where they are worn) **Answer: C**

3. CROWD is to DISPERSED as MEETING is to:
   A. RECESSED C. DEPRESSED
   B. ADJOURNED D. DISCUSSED
   Clues: A **crowd** is a large number of people **collected or gathered** together. **Dispersed** means to move a part or go away in different directions. These two words are opposite in meaning (crowd ≠ dispersed)
   Meeting is opposite with **recessed.**
   (**Meeting** = coming together to discuss; **recessed** not coming together) → (**antonym** relationship) **Answer: A**

4. FOOT is to LEG as HAND is to:
   A. FINGERS B. ARM C. THUMB D. TOE
   Clues: **Foot** is the lower part of **leg** as **hand** is the lower end part of **arm**. →(part to the whole relationship) **Answer: A**

5. SUCCESS is to EFFORT as FAILURE is to:
   A. LAZINESS C. DISASTER
   B. DEVOTION D. CHANCE
   Clues: **Success** is the result of **effort** as **failure** is the result of **laziness** →(cause – effect relationship). **Answer: A**

6. BEGIN is to START as EMPLOY is to:
   A. CELEBRATE C. END
   B. FIRE D. HIRE

Clues: **Begin** and **start** are similar words; **employ** and **hire** are synonyms, too. →(**Synonym relationship**). **Answer: D**

7. BEEF is OXEN as PORK is to:
   A. LION B. GOAT C. PIG D. SHEEP
   Clues: **Beef** is meat prepared from **oxen** as **pork** is meat prepared from a **pig**. → (The name of meat prepared from different animals). **Answer: C**

DIRECTIONS: Questions 8 – 10 consist of words printed in capital letters followed by four words. Choose the word that is most nearly **similar** in meaning to the word in capital letters. Then, find the question number and **blacken** the letter of your choice on the separate answer sheet provided.

8. MISERABLE:
   A. Abject B. Refutable C. Violet D. Challenging
   Answer: A

9. POVERTY:
   A. Deficit B. Humility C. Restraint D. Destitution
   Clues: Destitution is much more related with poverty. **Destituion** means being without money, food and other things necessary for life. **Answer: D**

10. MISFORTUNE:
    A. Collapse C. Adversity
    B. Crisis D. Defenselessness
    Clues: **Misfortune** means **bad luck** = **negative and unpleasant.** **Answer: C**

Antonyms

DIRECTIONS: Questions 11 – 13 consist of words printed in capital letters followed by four words. Choose the word that is most nearly **opposite** meaning to the word in capital letters. Then, find the question number and **blacken** the letter of your choice on the separate answer sheet provided.

11. MEAN:
    A. Cruel B. Friendly C. Malicious D. Unkind
    Clues: **Mean** (adj) which means **unkind** (of behavior).
    →The **opposite** is sociable or **friendly**. **Answer: B**

12. CONDENSE:
    A. Block B. Convert C. Summarize D. Expand

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Clues:
Condense (v) which means to put something such as a piece of writing into **fewer words** to put a lot of information into a small space. →The opposite word is **enlarge or expand**.

13. SELFISH:
Answer: D
A. Altruistic B. Meager C. Generous D. Greedy
Clues: **Selfish** (adj) means greedy. The antonym word is **given freely/generous**.
Answer: C

DIRECTIONS: Questions **14 – 18** deal with restoring words correctly into a passage. The following five words listed at random are removed from the passage given below. Numbers are written instead of the words. Read the passage and choose the word that correctly completes the passage. Then, **blacken** the letter of your choice on the separate answer sheet provided.

emissions exception unprecedented warmest trend

The Earth's climate has changed throughout history. The current warming __1__ is of particular significance because most of it is extremely likely to be the result of human activity since the mid-20th century and proceeding at a rate that is __2__. The planet's average surface temperature has risen about 1.1 degrees Celsius since the late 19th century, a change driven largely by increased carbon dioxide and other human-made __3__ into the atmosphere. Most of the warming occurred in the past 35 years, with 16 of the 17 warmest years on record occurring since 2001. Not only was 2016 the __4__ year on record, but eight of the 12 months that make up the year from January through September, with the __5__ of June – were the warmest on record for those respective months.

14. Which of the following is the correct word that should replace 3 (the third blank in the passage) is _____.
A. trend B. emissions C. warmest D. unprecedented
Answer: B

15. Which of the following is the correct word that should replace 5 (the fifth blank in the passage) is _____.
A. Exception B. trend C. warmest D. unprecedented
Answer: A

16. Which of the following is the correct word that should replace 2 (the second blank in the passage) is _____.
A. Exception B. emissions C. unprecedented D. warmest
Answer: C

17. The correct word that should replace 1 (the first blank in the passage) is _____.
A. trend B. emissions C. warmest D. unprecedented
Answer: A

18. Which of the following is the correct word that should replace 4 (the fourth blank in the passage) is _____.
A. exception B. trend C. unprecedented D. warmest
Answer: D

DIRECTIONS: In questions **19 – 23** three of the words are in the same classification, the remaining one word does not belong to the others. From the given four alternatives, choose the word that does not belong in the same classification as the others and **blacken** the letter of your choice on the separate answer sheet provided.

19. Choose the odd one out of the following words.
A. Anger C. August
B. Disgust D. Laughter
Clues: Words in option A, B and D are about sense of emotional feeling related to pitiness and excitement August in among months of the year.
Answer: C

20. Which one of the following words does **NOT** belong with the others?
A. Earth C. Mars
B. Cosmos D. Jupiter
Clues: Earth, Mars and Jupiter are planets found in the solar system.
Answer: B

21. Choose the word that is different from the other three.
A. Zoology C. Psychology
B. Ecology D. Biology
Clues: **Biology, Ecology** and **Zoology** are the scientific study of living things (animals and planets) their structure, relationship, etc.
Psychology: The scientific study of the mind and how it is influences behavior.
Answer: D

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22. Which one of the following words does **NOT** belong with the others?
A. Grape C. Orange
B. Apple D. Sugarcane
Clues: Grape, apple and orange are kinds of fruits. **Answer: D**

23. Choose the word that is different from the other three.
A. Leaf C. Stem
B. Dirt D. Root
Clues: **Leaf, stem** and **root** are parts of a tree – top to bottom
Dirt (N) is soil or earth. **Answer: B**

DIRECTIONS: Answer questions **24 – 26** by selecting the choice that contains the word or pair of words that correctly complete the statements. Then, **blacken** the letter of your choice on the separate answer sheet provided.

24. The natural balance between prey and predator has been increasingly _____, usually because of the human provided.
A. Recognized C. Disturbed
B. Predicted D. Observed
Answer: C

25. Typically, legal proceedings begin with an investigation meant to _____ evidence, followed by a trial that establishes whether that evidence _____ a criterion – "**beyond a reasonable doubt**" in criminal proceedings, and "**according to the preponderance of evidence**" in civil proceedings.
A. Verify: contradicts C. Show: declare
B. Collect: meets D. Mean: amounts
Answer: B

26. It was a war the prime minster and her wise counselors wished to _____ if they could and were determined in any event to _____ as long as possible. They were repeatedly meeting and discussing on this issue until the case was under their control.
A. Provoke: delay C. Avoid: postpone
B. Oppose: fight D. Curtail: Promote

Answer:

Logical and Analytical reasoning (27 – 35)

DIRECTIONS: In this section (Questions 27 – 32) each equation or a group of questions in based on a passage or set of conditions. Read each question carefully and select the best answer and **blacken** the letter of your choice on the separate answer sheet provided.

27. Today's secondary school students are not being educated. They are being trained. Their teachers demand little of them other than that they memorize facts and follow directions. The current emphasis on training in basic maths and verbal skills, while a useful step, rarely leads to the essential second step: development of independent critical thinking.
The author would probably consider which of the following aspects of a student's term paper to be most praiseworthy?
A. Evidence of extensive research prior to writing
B. Evidence of original insights and freshly developed concepts
C. The use of grammatically correct sentence structure
D. The choice of a challenging topic
Answer: B

28. Obesity is a serious problem in many developed countries. Research suggests that obesity can lead to a number of health problems including diabetes, asthma, and heart disease. Recent research has even indicated that there may be a relationship between obesity and some types of cancer. Major public health campaigns that increase awareness and propose simple lifestyle changes that will, with diligence and desire, eliminate or at least mitigate the incidence of obesity are a crucial first step in battling this critical problem.
This paragraph best supports the statement that
A. A decline in obesity would radically decrease the incidence of asthma.
B. People in those countries watch too much television and do not exercise enough.
C. Fast-food restaurants and unhealthy school lunches contribute greatly to obesity.

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D. Public health campaigns that raise consciousness and propose lifestyle changes are a productive way to fight obesity.

Answer: D

29. Alula must be a basketball player; he is wearing a basketball jersey. The conclusion above is valid only if it is true that:
A. All basketball players were basketball jerseys.
B. Only basketball players wear basketball jerseys.
C. Basketball players often wear basketball jerseys.
D. Basketball players are required to wear basketball jerseys.

Answer: C

30. A company worker has six children. No one else in the factory has six children. Which one of the following conclusions logically follows the statements given above?
A. All workers in the company have six children each.
B. Some of the company workers have more than six children.
C. Some of the company workers have more than six children.
D. Everybody in the company has children.

Answer: C

31. Today's secondary school students spend too much time thinking about trivial and distracting matters such as fashion. Additionally, they often dress inappropriately on college grounds. Rather than spending time writing another detailed dress policy, we should make school uniforms mandatory. If students were required to wear uniforms, it would increase a sense of discipline in our students. Another positive effect would be that teachers and administers would no longer have to act as clothing police, freeing them up to focus on more important issues. This paragraph best supports the statement that
A. Students are not interested in being part of a community.
B. Students who wear school uniforms get into better colleges.
C. School uniforms should be compulsory for secondary school students.
D. Inappropriate clothing leads to failing grades.

Answer: C

A. 3 B. 9 C. 12 D. 6

Solution: Let x – be the unknown number then $x + 3 = 2x - 3 \Rightarrow 2x - x = 3 + 3 \Rightarrow x = 6$

Answer: D

10. Three professors contested an election for a university president and received 1,136,7636 and 11,628 votes, respectively. What percentage of the total votes did the winning professor get?
A. 65% B. 57% C. 59% D. 61%

Solution: total participant = 1136 + 7636 + 11,628 = 20,400

winner get = 11628

Percentage of winning professor get $= \frac{11,628}{20,400} \times 100\%$

Percentage of winning professor get 57%

Answer: B

11. If one fourth of one – third of a number is 15, then what is three – tenth of that number?
A. 39 B. 45 C. 54 D. 34

Solution: one fourth of one third of the number is 15

Let x – be the unknown number

$\Rightarrow \frac{1}{4}$ of $\frac{1}{3}$ of $x = 15$

$\left(\frac{1}{4} \times \frac{1}{3} \times x\right) = 15 \Rightarrow x = 15 \times 12 = 180$

Three tenth of $x = \frac{3}{10}$ of $180 = \frac{3}{10} \times 180 = 54$

Answer: C

12. An athlete completes a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. What is the total journey in km?
A. 216 B. 225 C. 218 D. 230

Solution: $v_1 = 21$ km/hr

$V_2 = 24$ km/h

$\Delta t = 10$ hr

$S_{total} = v_{av} \times \Delta t$

$S_{total} = \left(\frac{v_1 + v_2}{2}\right)\Delta t$

$S_{total} = \frac{(21\text{km/hr} + 24\text{km/hr})}{2} 10\text{hr} = 225\text{km}$

Answer: B

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13. If $y \neq 0$ and $xy = \frac{y}{6}$, what is the value of x?

A. $\frac{1}{2}$ B. $\frac{1}{8}$ C. $\frac{1}{4}$ D. $\frac{1}{6}$

Solution: $xy = \frac{y}{6} \Rightarrow \frac{xy}{y} = \frac{y}{6y} \Rightarrow x = \frac{1}{6}$

Answer: D

14. What number should come next in the series: 7, 10, 8, 11, 9, 12, …?

A. 13 B. 10 C. 12 D. 7

Solution: 7, 10, 8, 11, 9, 12, y …

$1^{st}, 3^{rd}, 5^{th}$ … 7, 8, 9, … is an arithmetic sequence with common difference 1

$2^{nd}, 4^{th}, 6^{th}, \ldots = 10, 11, 12, \ldots$

$\therefore y = 10$ hence 7, 10, 8, 11, 9, 12, 10, …

Answer: B

15. The difference between a positive improper fraction and its reciprocal is $\frac{9}{20}$? What is the value of the fraction?

A. $\frac{5}{4}$ B. $\frac{3}{5}$ C. $\frac{4}{3}$ D. $\frac{4}{2}$

Solution: $\frac{a}{b} = \frac{5}{4}$ is a fraction and $\frac{b}{a} = \frac{4}{5}$ is proper fraction then

$\frac{a}{b} - \frac{b}{a} = \frac{a^2 - b^2}{ab} = \frac{25-16}{20} = \frac{9}{20}$

Answer: A

16. What is the product of $(\sqrt{7}-2)(\sqrt{7}+2)$?

A. 5 B. 4 C. 2 D. 3

Solution: $(\sqrt{7}-2)(\sqrt{7}+2) = 7 + 2\sqrt{7} - 2\sqrt{7} - 4 = 3$

Answer: D

17. The sum of ages of 5 children born at the interval of 3 years each is 50 years. What is the age of the youngest child?

A. 8 years B. 4 years C. 6 years D. 10 years

Solution: Let x – be the age of the youngest child x, x + 3, x + 6, x + 9, x + 12 patterns of the age of 5 children i.e.,

$x + x + 3 + x + 6 + x + 9 + x + 12 = 50 \Rightarrow 5x + 30 = 50 \Rightarrow x = 4$

∴ 4, 7, 10, 13, 16 the age of youngest child 4 years

Answer: B

18. What number should come next in the series 544, 509, 474, …?

A. 445 B. 420 C. 404 D. 414

Solution: 544, 509, 474, 439, … forms decreasing arithmetic sequence with common difference d = -35, hence the next term is 404.

Answer: C

19. A library has an average of 510 visitors on Sundays and 240 on other days. What is the average number of visitors per day in a month of 30 days beginning with a Sunday?

A. 285 B. 283 C. 276 D. 290

Solution: The average number of visitors per day.

$\bar{x} = \frac{f_1\bar{x}_1 + f_2\bar{x}_2}{f_1 + f_2}$ where f_1 – number of Sundays on months

$\bar{x}_1$ - average of Sunday visitors

f_2 – number of other days

$\bar{x}_2$ - average of other days

$\bar{x} = \frac{4\times510 + 26\times240}{4+26} = \frac{2040+6240}{30} = \frac{8280}{30} = 276$ per day

Answer: C

20. What number should come next in the series: 58, 52, 46, 40, 34, …

A. 32 B. 30 C. 28 D. 34

Solution: The given pattern 58, 52, 46, 40, 34, … form arithmetic sequence with common difference d = -6

Then the sixth term $a_6 = 34 + (-6) = 28$

Answer: C

Data interpretation

Table: Number of candidates that took and qualified in a license examination from different regions over the year? Question (21 – 25) based on the following table.

Region	year									
	1997- 2000		1998 – 2001		1999 - 2002		2000 - 2003		2001 – 2004	
	Took	Qual	Took	Qual	Took	Qual	Took	Qual	Took	Qual
AA	5200	720	8500	980	7400	850	6800	775	9500	1125
AM	7500	840	9200	1050	8450	920	9200	980	8800	1020
OR	6400	780	8800	1020	7800	890	8750	1010	9750	1250
TI	8100	950	9500	1240	8700	980	9700	1200	8950	995
SP	7800	870	7600	940	9800	1350	7600	945	7990	885

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121. What is the percentage of total number of qualified candidates to the total number of candidates took the exam among all the five regions in 2002?

A. 11.84% B. 11.52% C. 11.48% D. 11.89%

Solution: Percentage of qualified candidates

$$= \frac{4990}{42,150} \times 100\% = 11.84\%$$

Answer: A

22. What is the percentage of candidates qualified from region AM for all the years together, over the candidate took the exam from region AM during all the years together?

A. 11.75% B. 11.46% C. 11.57% D. 11.15%

Solution: Percentage of candidates qualified from region AM all the years together.

$$= \left(\frac{840+1050+920+980+1020}{7500+9200+8450+9200+8800} \right) 100\% = \frac{4,810}{43,150} \times 100\% = 11.15\%$$

Answer: D

23. In which of the given years the number of candidates took the exam from region OR has maximum percentage of qualified candidate?

A. 2001 B. 2003 C. 2004 D. 2000

Solution: Qualified parentage of OR candidates took exam

$$\text{In 2000} = \frac{780}{6400} \times 100\% = 12.19\%$$

$$\text{In 2001} = \frac{1020}{8800} \times 100\% = 11.591\%$$

$$\text{In 2003} = \frac{1010}{8750} \times 100\% = 11.543\%$$

$$\text{In 2004} = \frac{1250}{9750} \times 200\% = 12.82\%$$

Maximum percentage qualified in 2004 **Answer: C**

24. What is the average number of candidates who took the exam from region Tl during the given year?

A. 8800 B. 8900 C. 8860 D. 8990

Solution: Average $= \frac{8100+9500+8700+9700+8950}{5}$

$$\text{Average} = \frac{44,950}{5} = 8,990$$

Answer: D

25. What percent is the total number of qualified candidates in 1997 – 2000 to the total number of those qualified 2001 – 2004?

A. 84% B. 76% C. 78% D. 80%

Solution:

In 1997–2000 qualified candidates = 720+840+780+950+870= 4160

In 2001 – 2004 qualified candidates = 1125 + 1020 + 1250 + 995 + 885 = 5275

Percent of total candidates in 1997 – 2000 to 2001 – 2004

$$= \frac{4160}{5275} \times 100\% = 78\%$$

Answer: C

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1. If n is an integer such that $(-3)^{2n} = 3^{9-n}$ then what si the value of n.
   A. 4 B. 3 C. 2 D. 1
   Solution: Here at $(-3)^{2n} = 9^n$, so that $9^n = 3^{9-n} \Leftrightarrow 3^{2n} = 3^{9-n}$
   $\Rightarrow 2n = 9 - n \Rightarrow 3n = 9$, therefore, $n = \frac{9}{3} = 3$ **Answer: 3**

2. What is the value of 5.005/2.0002?
   A. 2.05 B. 2.501 C. 2.50 D. 2.5015
   Solution: Here at we have:
   $\frac{5.005}{2.002} = \frac{5005 \times 10^{-3}}{2002 \times 10^{-3}} = \frac{5005}{2002} = 2.50$ **Answer: C**

3. How many multiple of 4 are their between 12 and 96 inclusive?
   A. 22 B. 23 C. 24 D. 25
   Solution: Here we can use arithmetic progration concept.
   $A_1 = 12, d = 4, A_n = 96$
   $A_n = A_1 + (n-1)d \Rightarrow 96 = 12 + (n-1)4$
   $\Rightarrow 4n - 4 + 12 = 96$ implies $4n = 88$
   $n = 8\frac{8}{4} = 22$. **Answer: A**

4. If the product of six integers in negative, at most how many of the integers can be negative?
   A. 2 B. 3 C. 4 D. 5
   Hint: From six given any integers let one of them is positive and the rest 5 of them are negative.
   - (5 positive)(1 negative) = (+)(+)(+)(+)(+).(−) = negative
   - (3 positive) (3 negative) = (+)(+)(+).(−)(−)(−) = negative
   - (1 positive)(5 negative) = (+).(−)(−)(−)(−)(−) = positive

   Answer: D

5. Which one of the following is the largest?
   A. 80% of 340 C. 60% of 450
   B. 30% of 900 D. 50% of 540
   Hint: x% of $y = \frac{(x\%)(y)}{100\%}$, here at:
   A. $\frac{(80\%)(340)}{100\%} = 272$ C. $\frac{60\% \times 340}{100\%} = 270$
   B. $\frac{30\% \times 900}{100\%} = 270$ D. $\frac{50\% \times 540}{100\%} = 270$ **Answer: A**

6. $\left(\frac{1}{5}\right)^5$ is equal to _____
   A. 0.03125 B. 0.00625 C. 0.00025 D. 0.00032
   Hint: Here at $\left(\frac{1}{5}\right)^5 = (0.2)^5 = (2 \times 10^{-1})^5 = 2^5 \times 10^{-5}$
   $\therefore \left(\frac{1}{5}\right)^5 = 32 \times 10^{-5} = 0.00032$

7. By how much is $\frac{3}{7}$ large than 20% of 2.
   A. $\frac{1}{35}$ B. $\frac{4}{7}$ C. $\frac{3}{7}$ D. $\frac{2}{35}$
   Solution: 20% of $2 = \frac{20\% \times 2}{100\%} = \frac{40}{100} = 0.4 = \frac{2}{5}$
   $\frac{3}{7} - \frac{2}{5} = \frac{15-14}{35} = \frac{1}{35}$
   $\therefore \frac{3}{7}$ is larger than 20% of 2 by $\frac{1}{35}$.

8. What is x-percent of y divided by y percent of x?
   A. x B. 1 C. y D. 5
   Hint: x% of $y = \frac{(x\%)(y)}{100\%}$
   y% of $x = \frac{(y\%)(x)}{100\%}$
   $\frac{x\% \text{ of } y}{y\% \text{ of } y} = \frac{\frac{(x\%)(y)}{100\%}}{\frac{(y\%)(x)}{100\%}} = \frac{(x\%)(y)}{(y\%)(x)} = \frac{(xy)\%}{(xy)\%} = 1$

Logical Reasoning

9. Consider the positions of x, x^2 and x^3 on the number line. Which of the following could be a value of x.
   A. 2 B. $\frac{5}{3}$ C. $\frac{-2}{5}$ D. $\frac{-5}{2}$

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Hint: On the number line, the smallest number are located to the left of the other. Here at: $x < x^3 < x^2$.

$\Rightarrow -\frac{2}{5} < \left(\frac{-2}{5}\right)^3 < \left(\frac{-2}{5}\right) = -0.4 < -0.064 < 0.16$ **Answer: C**

10. If $x = 17 \times 21$, which one of the following is NOT an integer.
A. $x \div 3$ B. $x \div 7$ C. $x \div 42$ D. $x \div 5$
Solution: Hint: $x = 17 \times 21 = 17 \times 7 \times 3 = 51 \times 7$;
$x = 17 \times 21$ is divisible by 3, 7 and 51. **Answer: C**

11. If A and B are positive integers and $A\Delta B = A(A + 2B)$, what is the value of $2\Delta[4\Delta(1\Delta3)]$?
A. 40 B. 64 C. 88 D. 164
Hint: $A\Delta B = A(A + 2B)$
$2\Delta[4\Delta(1\Delta3)] = 2\Delta[4\Delta(1(1+2(3))] = 2\Delta(4\Delta7)$
$\Rightarrow 2\Delta(4\Delta7) = 2\Delta[4(4 + 2(7))] = 2\Delta72$
$\Rightarrow 2\Delta72 = 2(2 + 2(72)) = 2(146) = 292$
No answer

12. How many numbers between 100 and 300 meet one or both of the condition given in the two statements below?
Statement 1: The number begins with 2.
Statement 2: The number ends with 2.
A. 90 B. 100 C. 110 D. 120
Explanation: Here at:
The number begins with 2 are 200, 201, 202, . . . 299.
Using arithmetic sequence: $A_1 = 200$, $d = 1$
$A_n = A_1 + (n - 1)d \Leftrightarrow 299 = 200 + (n - 1) \Rightarrow n - 1 = 99 \Rightarrow n = 100$
And the number ends with 2 are: 102, 112, 122, . . ., 292, so that
$a_1 = 102$, $d = 10$, $a_n = a_1 + (n - 1)d \Leftrightarrow 292 = 102 + (n - 1)10$
$\Rightarrow 10n - 10 = 292 - 102 \Leftrightarrow 10n = 200$, gives $n = 20$
Therefore, the total number of the two condition is
$100 + 20 = 120$.

Quantitative Reasoning

13. In which of the following is the first number or expression equal to the second number or expression.
A. $\left[\frac{1}{3}+\frac{1}{4}\right] / \left[\frac{1}{3}-\frac{1}{4}\right]; 7$ C. $25(26) + 26(75); 2500$

B. $8 - (-12) - 5; 5 + (-8) + 12$ D. $\left[5-\frac{1}{10}\right] / \left[10+\frac{1}{10}\right]; \frac{1}{2}$

Answer: A

14. If x and y are integers and $x > y > 0$, how many integers are there between, but not including x and y?
A. $x - y$ B. $x + y$ C. $x - y - 1$ D. $x + y - 1$
Explanation:
The number of positive integer between x and y, excluding x and y for $x > y$ is $x - y - 1$.
Clue: The number of integer between
i) 4 and 3 is $4 - 3 - 1 = 0$ (No integer)
ii) 4 and 2 is $4 - 2 - 1 = 1$ (one integer that is 3)
iii) 8 and 3 is $8 - 3 - 1 = 4$
iv) x and y is $x - y - 1$ **Answer: C**

15. The closest value of the product of 408 and 500 is _____.
A. 20,000 B. 40,000 C. 200,000 D. 400,000
Hint: $408 \sim 400$, so that
$408 \times 500 \sim (400)(500) = 200{,}000$

16. Which of the following numbers does not satisfy the inequality $x - 3 > 3x - 2$?
A. -3 B. -1 C. 0 D. -2
Solution: $x - 3 > 3x - 2 \Leftrightarrow -3 + 2 > 3x - x$

$\Rightarrow -1 > 2x$ implies $x < -\frac{1}{2}$

The set of all real number $\left\{x : x < -\frac{1}{2}\right\}$ satisfy the inequality.

Answer: C

17. Divide 64 by a quarter and add 5. What is the resulting number?
A. 261 B. 256 C. 21 D. 11

Solution: $64 \div \frac{1}{4} + 5 = 64 \times 4 + 5 = 261$

18. Ali, Aminat, and Yosef have their own collections of stamps. If the ratio of Ali's stamps to those of Aminat is 5 to 1 and the ratio of Aminat's stamps to those of Yosef is 3 to 2, what si the ratio of Ali's to Yosef's stamps?
A. 5: 1 B. 5: 2 C. 10:3 D. 15:2

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Here is how to crack it

Let Ali = a, Aminat = b and Yosef = C a:b = 5: 1 and b : c = 3: 2

$$\frac{a}{c}=\left(\frac{a}{b}\right)\left(\frac{b}{c}\right)=\left(\frac{5}{1}\right)\left(\frac{3}{2}\right)=\frac{15}{2}=15:2$$

19. Which one of the following sum is Largest?

A. $\frac{5}{6}+\frac{7}{17}$ B. $\frac{6}{7}+\frac{5}{11}$ C. $\frac{7}{8}+\frac{3}{7}$ D. $\frac{8}{9}+\frac{1}{3}$

Here is how to find (identify)

$\frac{5}{6}<\frac{6}{7}<\frac{7}{8}<\frac{8}{9}$ and $\frac{1}{3}<\frac{7}{17}<\frac{3}{7}<\frac{5}{11}$

$\Rightarrow \frac{6}{7}+\frac{5}{11}=\frac{66+35}{77}=\frac{101}{77}$ is largest

20. $\frac{5^6-5^5}{4}$ is equal to:

A. $\frac{1}{4}$ B. $\frac{5}{4}$ C. 5^5 D. 5

Here at: $\frac{5^6-5^5}{4}=\frac{5^5(5-1)}{4}=\frac{5^5(4)}{4}=5^5$

21. In the sequence -24, -18, -12, -6; . . . each number after the first is 6 greater than the preceding number. Which of the following could NOT be a number in the sequence?

A. 648 B. 762 C. 963 D. 1104

Solution: Here is how to identify -24, -18, -12, -6, . . . are arithmetic sequence with common difference of d = 6, so the general n^{th} term of sequence is $\boxed{A_n = A_1 + (n-1)d}$.

$A_n = -24 + (n-1)(6) = 6n - 30$

All the terms can be generated by $6n - 30$ where, n = 1, 2, 3, . . .

Here at:

A. $6n - 30 = 648 \Rightarrow 6n = 678$, therefore, $n = \frac{678}{6} = 113$

B. $6n - 30 = 762 \Rightarrow 6n = 792$, $\therefore n = \frac{792}{6} = 132$

C. $6n - 30 = 963 \Rightarrow 6n = 993 \Rightarrow n = 165.5$

Since n = 165.5 is not natural number.

$\therefore$ 963 is not part of sequence.

22. Suppose that $-1 \le x \le 0$.

Quantity A	Quantity B
$x^3 + 1$	$x^2 + 2x + 2$

A. Quantity A is greater than quantity B.
B. Quantity A is less than quantity B.
C. The two quantities are equal.
D. The relationship cannot be determined from the information given.

Here is how to crack it

Let's start with $x = -1, -\frac{1}{4}, -\frac{1}{2}, 0$

	Quantity A	Quantity B
x = -1	$(-1)^3 + 1 = 0$	$(-1)^2 + 2(-1) + 2 = 1$
$x = -\frac{1}{4}$	$\left(-\frac{1}{4}\right)^3 + 1 = \frac{63}{64}$	$\left(-\frac{1}{4}\right)^2 + 2\left(-\frac{1}{4}\right) + 2 = \frac{25}{16}$
$x = -\frac{1}{2}$	$\left(-\frac{1}{2}\right)^3 + 1 = \frac{7}{8}$	$\left(-\frac{1}{2}\right)^2 + 2\left(-\frac{1}{2}\right) + 2 = \frac{5}{4}$

(Quantity A) < (Quantity B) **Answer: B**

23. Consider the following two quantities.

Quantity A	Quantity B
$2^{28} + 2^{28} + 2^{28}$	2^{29}

A. Quantity A is greater than quantity B.
B. Quantity A is less than quantity B.
C. The two quantities are equal.
D. The relationship cannot be determined from the information given.

Here is how to crack it.

Quantity A $= 2^{28} + 2^{28} + 2^{28} = 2^{28}(1 + 1 + 1) = 3(2^{28})$

Quantity B $= 2^{29} = (2)(2^{28})$

$\Rightarrow 3(2^{28}) > 2(2^{28})$ **Answer: A**

24. Two sides of a right triangle are 10 and 12 unit long. Which of the following could be the length of the third side?

A. $2\sqrt{11}$ B. $3\sqrt{7}$ C. $3\sqrt{11}$ D. A and B

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Hint: Let the hypotenuse of triangle be C = 12 unit.

$C^2 + 10^2 = 12^2 \Rightarrow C^2 = 144 - 100 = 44$

$C = \sqrt{44} = 2\sqrt{11}$ unit

25. If x and y are prime numbers greater than 2, which of the following must be true?

(I) x = y is even

(II) xy is odd

(III) $x^2 - y^2$ is even

A. I and II only C. II and III only

B. I and III only D. I, II and III

Clue: The set of prime numbers are 3, 5, 7, . . .

$11 + 5 = 16 \leftarrow$ even

$(11)(5) = 55 \leftarrow$ odd

$11^2 - 5^2 = 121 - 25 = 96 \leftarrow$ even **Answer: D**

26. On a test containing 60 questions, Azeb answered 75% of the first 40 questions correctly. What percent of the other 20 questions does she need to answer correctly for her grade on the entire examination to be 80%.

Here is the way how to find.

Let x be the required (75% of 40) + (x% of 20) = 80% of 60.

$$\Rightarrow \left(\frac{75}{100}\times 40\right) + \frac{x}{100}(20) = \left(\frac{80}{100}\right)(60)$$

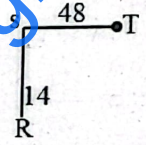
$$\Rightarrow 30 + 0.2x = 48$$

$$\Rightarrow 0.2x = 48 - 30 = 18 \Leftrightarrow x = \frac{18}{0.2} = 90\%$$

27. If an airplane starts at point R and travels 14 miles directly north to S, then 48 miles directly east to T, what is the straight line distance (in miles) from T to R?

A. 25 B. 34 C. 50 D. 2500

Here is how to crack it.



$(TR)^2 = (14)^2 + (48)^2 = 2500$

$TR = \sqrt{2500} = 50$ miles

28. A circular region has circumference **C** centimeters and area **k** square centimeters. If C = 3k. What is the radius of the circle in centimeters?

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{3}{2}$ D. 3

Here is how to crack it.

$C = 2\pi r$ and $k = \pi r^2$

$$\Rightarrow r = \frac{C}{2\pi} \text{ and } r = \sqrt{\frac{k}{\pi}} \text{ implies } \frac{C}{2\pi} = \sqrt{\frac{k}{\pi}}$$

Since C = 3k implies $\frac{3k}{2\pi} = \sqrt{\frac{k}{\pi}}$ squaring

$$\frac{k}{\pi} = \frac{9k^2}{4\pi^2} \Leftrightarrow \frac{9k^2}{k} = \frac{4\pi^2}{\pi} \Rightarrow k = \frac{4\pi}{9}$$

Therefore, $r = \sqrt{\frac{k}{\pi}} = \sqrt{\left(\frac{4\pi}{9}\right)\frac{1}{\pi}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$.

29. A train travels 60 miles per hour for 3 hours and then 45 miles per hours for 2 hours. What is the trains average speed in miles per hour during the 5 hour period?

A. 55 B. 54 C. 51 D. 50

Solution:

The distance travel in first 3 hours:

$d_1 = (60 \text{ mi/hr})(3 \text{ hr}) = 180$ mile

The distance travel in 2nd two hours is:

$$d = \left(\frac{45\text{mil}}{\text{hr}}\right)(2\text{hr}) = 90\text{mile}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Time taken}} = \frac{180+90}{5} = 54$$ **Answer: B**

30. If m is a number between 0 and 1, which of the following is NOT more than m?

A. m_2 B. $\frac{1}{m}$ C. 2m D. $\sqrt{m}$

Clue: Let $m = \frac{1}{2}$

A. $m^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ C. $2m = 2\left(\frac{1}{2}\right)$

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B. $\frac{1}{m} = \frac{1}{\frac{1}{2}} = 2$ **Answer: A**

31. If $\frac{1}{5}$ of a certain number is 6 then $\frac{1}{6}$ of the number is _____.

A. 5 B. 6 C. 30 D. 36

Solution: Here at, let a certain number is x.

$\Rightarrow \frac{1}{5}x = 6 \Rightarrow x = 30$ and $\frac{1}{6}x = \frac{1}{6}(30) = 5$ **Answer: A**

Data Interpretation

Number of car Accidents in X Town and Estimates of property Damage in Ethiopian Birr (ETB).

Year	Number of car Accidents	Property damage estimate in ETB
1990	1270	12,050,000
1991	1289	12,688,000
1992	1345	12,090,000
1993	1578	12,100,000
1994	1645	12,400,000
1995	1888	12,500,000
1996	2010	12,700,000
1997	2100	12,800,000
1998	2150	12,950,000
1999	2175	13,100,000
Total	17,450	125,378,000

32. For which specific year is the ratio of property damage estimate to number of car accidents the highest?

A. 1999 B.. 1998 C. 1991 D. 1990

Solution:

In 1999: $\frac{13,100,000}{2,175} = 6023$

In 1998: $\frac{12,950,000}{2150} = 6023$

In 1991: $\frac{12,688,000}{1289} = 9843$

In 1990: $\frac{12,050,000}{1270} = 9488$ **Answer: A**

33. Which of the following can be inferred from the data in the table above?

I. The increase in car accidents from 1990 to 1991 is the lowest during the 10 years period.

II. The increase in car accidents from 1994 to 1995 is the highest during the 10 year period.

III. On average, one car accident is estimated to cause about 9488 ETB is property damage during the 10 – year period.

A. I and II B. I and III C. I, II and III

Solution:

I) $1289 - 1270 = 19 \leftarrow$ lowest car accident increase

II) $1888 - 1645 = 243 \leftarrow$ highest car accident increase

III) Average car accident to cause property damage = $\frac{12,378,000}{17,450} = 7184$ **Answer: A**

34. What is the annual average (arithmetic mean) of car accidents recorded in town (x) from 1990 to 1999?

A. 3490 B. 1745 C. 1163 D. 8725

Hint:

$$\text{Average} = \frac{\text{Total number of car accidents}}{10} = \frac{17,450}{10} = 1745$$

Word problem

35. Meron is now 14 years older than Nati. If in 10 years Meron will be twice as old as Nati, how old will Meron be in 5 years?

A. 29 B. 21 C. 23 D. 33

Solution:

Let x = age of Nati

y = age of Meron

$\Rightarrow y = x + 14 \leftarrow$ The present age of Meron

In 10 years meron's age will be:

$y + 10 = 2(x + 10) = 2x + 20$

$y + 10 = 2x + 20$, but $y = x + 14$

$\Rightarrow (x + 14) + 10 = 2x + 20$

$\Rightarrow 24 - 20 = 2x - x = x$

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$\Rightarrow x = 4$, $y = 18 \leftarrow$ present age of Nati and Meron

In 5 years will be $x = 4 + 5$ and $y = 18 + 5 = 23$. **Answer: C**

36. When 10 is divided by the positive integer P_1 then the remainder is $P - 4$. Which of the following could be the value of P?

A. 8 B. 7 C. 4 D. 3

Clue: If 10 is divided by P, the remainder is $P - 4$.

A. When 10 is divided by 8. The remainder is 2.

$\Rightarrow P - 4 = 2 \Rightarrow P = 6$, but $P = 8$

B. When 10 is divided by 7, the reminder is 3.

$\Rightarrow P - 4 = 3 \Rightarrow P = 7$ is the same $7 - 4 = 3$ **Answer: B**

37. The average (arithmetic mean) of 6 numbers is 8.5. When one number is discarded, the average of the remaining numbers becomes 7.2. What is the discarded number?

A. 15.4 B. 15.0 C. 12.4 D. 10.0

Clue: Let a, b, c, d, e, f are the number

$$\Rightarrow \frac{a+b+c+c+d+e+f}{6} = 8.5 \Rightarrow a+b+c+d+e+f = 6(8.5) = 5$$

Let a = discarded number

$$\Rightarrow \frac{b+c+d+e+f}{5} = 7.2 \Rightarrow b+c+d+e+f = (7.2)(5) = 36$$

$\therefore (a+b+c+d+e+f) - (b+c+d+e+f) = 51 - 36 = 15$

Answer: B

38. A football club collected birr 599 from its members. If each member contributed at least birr 12, what is the greatest number of members the club could have?

A. 52 B. 51 C. 50 D. 49

Clue:

A. $\frac{599}{52} = 11.51$ birr/member

B. $\frac{599}{51} = 11.74$ birr/ member

C. $\frac{599}{50} = 11.98$ birr/member

D. $\frac{599}{49} = 12.22$ birr/member **Answer: D**

39. A lamp is manufactured to sell for birr 35, which yields a profit of 25% of cost. If the profit is to be reduced to 15% of cost, the new retail price will be:

A. 44 B. 32 C. 16 D. 12

Clue: Let x = cost price

$\Rightarrow x + 25\%$ of $x = 35 = 25\%$ of $x = 35 - x$

$$\Rightarrow \frac{(25)(x)}{100} = 35 - x \text{ implies } 25x = 3500 - 100x$$

$$\Rightarrow 125x = 3500 \text{ provides } x = \frac{3500}{125} = 28$$

If the profit is to be reduced to 15% of cost, the new selling price will be: $28 + 15\%$ of $28 = 28 + \frac{(15)(28)}{100} = 32.2$ birr **Answer: B**

40. If $n = 3$, what is the value of $2^{2n} + 1$.

A. 13 B. 17 C. 33 D. 65

Hint: $2^{2n} + 1 = (2^2)^n + 1 = 4^n + 1$

When $n = 3$, $4^3 + 1 = 64 + 1 = 65$

41. What is the greatest common factor of 42, 126 and 210?

A. 6 B. 14 C. 21 D. 42

Hint: Express 42, 126 and 210 are in prime factorizations.

$\rightarrow 42 = 2 \times 3 \times 7$

$\rightarrow 126 = 2 \times 3^2 \times 7$

$\rightarrow 210 = 2 \times 3 \times 5 \times 7$

$\rightarrow$The greatest common factor of 42, 126 and 210 is $2 \times 3 \times 7 = 42$.

42. If a radius of a circle inscribed in a square is k, then the perimeter of the square is:

A. K^2 B. $\frac{k}{4}$ C. $4k^2$ D. 8k

Hint: If radius, r = k, then diameter = 2k $\leftarrow$ the side length of the square.

Perimeter = (4)(2k) = 8k **Answer: D**

43. Which of the following fractions has the smallest value?

A. $\frac{2}{9}$ B. $\frac{1}{7}$ C. $\frac{1}{8}$ D. $\frac{3}{11}$

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Hint: Here at, $\frac{2}{9}=\frac{1}{4.5}$ and $\frac{3}{11}=\frac{1}{11/3}=\frac{1}{3.7}$

$\Rightarrow \frac{1}{8}<\frac{1}{7}<\frac{1}{4.5}<\frac{1}{3.7}=\frac{1}{8}<\frac{1}{7}<\frac{2}{9}<\frac{3}{11}$

Answer: C

44. If the sum of 30, 50 and y is 180, then the average (arithmetic mean) of 30, 50 and y is:

A. 60 B. 70 C. 80 D. 100

Hint: mean $(\bar{x})=\frac{30+50+y}{3}=\frac{180}{3}=60$

45. If $x>0$ and $2x-1=\frac{1}{2}(x+1)$, then $x=$ _____.

A. 1 B. $2\sqrt{2}$ C. $\frac{1}{2}$ D. $\sqrt{2}$

Hint: $2x-1=\frac{1}{2}x+\frac{1}{2}\Rightarrow 2x-\frac{1}{2}x=\frac{1}{2}+1$

$\Rightarrow \frac{3x}{2}=\frac{3}{2}\Leftrightarrow x=1$

46. If the value of the following expression $\left[\left(1+\frac{m}{n}\right)/\left(\frac{O}{P}\right)\right]$ is to be halved by doubling exactly one of the five numbers l, m, n, o or p which one should be doubled?

A. P B. O C. n D. m

Hint: The expression is equivalent to

$\left(1+\frac{m}{n}\right)\div\frac{O}{P}=\left(\frac{n+m}{n}\right).\left(\frac{P}{O}\right)$.

Doubling the letter 'O' the expression become halved.

Answer: B

47. If $y=\frac{a}{a+b}$ and $x=\frac{a}{b}$ what is y in terms of x?

A. $-\frac{1}{x}$ B. $1+x$ C. $1+\frac{1}{x}$ D. $\frac{x}{x+1}$

Hint: $x=\frac{a}{b}\Rightarrow a=bx$

Therefore, $y=\frac{a}{a+b}=\frac{bx}{bx+b}=\frac{bx}{b(x+1)}=\frac{x}{x+1}$

48. If the perimeter of a triangle is 18 then the length of one of sides **CANNOT** be:

A. 8 B. [illegible] C. 3 D. 1

Hint: Let a, b and c are length of sides of triangle. We have given $a+b+c=18$.

The following three cases must satisfy.

(i) $a+b>c$, (ii) $a+c>b$ (iii) $b+c>a$ thus

A. If $c=8$, $a+b>8$, but $a+b=10$

B. If $c=9$, $a+b>9$, but $a+b=9$ because $a+b+c=18$.
Therefore, $9>9$ is false.

C. If $c=3$, $a+b>3$, but $a+b=15$
$\therefore 15>3$ is true

D. If $c=1$, $a+b>1$, but $a+b=17$
$\therefore 17>1$ is true

Answer: B

49. Which of the following has the greatest value?

A. $\sqrt{0.3}$ B. 0.3 C. $\frac{1}{3}$ D. $\frac{2}{5}$

Hint: $\frac{1}{3}=0.333\ldots$ and $\frac{2}{5}=0.4$

Answer: D

50. $(C+x)/x$ is equal to:

A. C^2+1 B. c C. D. $\frac{c}{x}+x$

Hint: $\frac{c+x}{x}=\frac{c}{x}+\frac{x}{x}=\frac{c}{x}+1$

51. Which of the following numbers **Does not** have a reciprocal?

Hint: The reciprocal of x is $\frac{1}{x}, x\neq 0$.

A. 0 B. $\frac{1}{3}$ C. -2 D. 1

Answer: A

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52. Which one of the following expressions has an even integer value for all integers a and c?
A. 80 + 2ac B. 3a + 3c C. 2a + c D. a + 2c
Hint: Every even integers are expressed in multiple of 2. i.e. {0, 2, 4, 6, . . . 2n} where $n \in \mathbb{Z}$. Hence: $8a + 2ac = 2(4a + ac)$ are even integers for all a and c. **Answer: A**

53. If x = 3 is one solution to the equation $x^2 + rx - 20 = 4$ then r is equal to _____.
A. 5 B. -8 C. 3 D. 8
Hint: When x = 3; $(3)^2 + r(3) - 20 = 4$

$3r = 24 - 9 = 15$, therefore, $r = \frac{15}{3} = 5$

54. A triangle has a base (B) and an altitude (h). A second triangle has a base twice the altitude of the first triangle, and an altitude twice the base of the first triangle. What is the area of the second triangle?
A. $\frac{1}{2}Bh$ B. $\frac{1}{2}B^2h^2$ C. 2Bh D. 4Bh
Hint: First triangle base = B, and altitude = h
Second triangle base = twice of first triangle altitude = 2h
Second triangle altitude = twice of first triangle base (B) = 2B.

∴Area of second triangle = $\frac{1}{2}(2B)(2h) = 2Bh$. **Answer: C**

55. A certain in machine fills a bag with 7 kg of potato chips in 3.5 seconds. At this rate, how many seconds will it take the machine to fill a bag with 15 kg of potato chips?
A. 6.5 B. 7.0 C. 7.5 D. 8
Hint: Chips in kilogram per second = $\frac{7\text{kg}}{3.5\text{sec}} = 2\text{kg/sec}$
The time required to fill 15 kg of potato chips is
15 kg/ 2kg/sec = 7.5 sec.

56. A piece of paper with an area of 60 square centimeter is divided into two pieces so that the area of one is $\frac{2}{3}$ of the area of the other. What is the area of one of the pieces (in square centimeter)?
A. 20 B. 24 C. 30 D. 45

Hint: $\frac{2}{3}$ of 60 = 40 and $\frac{1}{3}$ of 60 = $\frac{1}{3}(60) = 20$ **Answer: A**

57. Sales for a business were 3 million dollars more the second year than the first, and sales for the third year were double the sales for the second year. If sales for the third year were 38 million dollars, what were sales in millions of dollars, for the first year?
A. 16 B. 17.5 C. 20.5 D. 22
Hint: Let sales for first year = A
Sales for 2nd year = B
Sales for 3rd year = C
B − A = 3,000,000 ⇒ B = 3,000,000 + A
C = 2B = 38,000,000
2B = 19,000,000
3,000,000 + A = 19,000,000 ⇒A = 16,000,000 **Answer: A**

58. In a group of 15, 7 can speak English, 8 can speak Arabic, and 3 can speak neither. What fraction of the group can speak both English and Arabic?
A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{5}$ D. $\frac{2}{3}$
Hint: Let who speak: English = A and Arabic = B
$n(A \cup B)' = 3$ but $n(A \cup B)' = n(\cup) - n(A \cup B)$ we have
$n(\cup) = 15$, so that $n(A \cup B) = n(\cup) - n(A \cup B)' = 15 - 3 = 12$
$\Rightarrow n(A \cap B) = n(A) + n(B) - n(A \cup B) = 7 + 8 - 12 = 3$
Therefore, a fraction of group can speak both English and Arabic are $\frac{3}{15} = \frac{1}{5}$. **Answer: B**

Data Interpretation
Questions 58 – 60 below are based on the hypothetical data given in table below.
From the table: Total population primary school age population and school environment by sex and regional in thousands.

Region	Total population		School age population		School environment	
	Male	Female	Boys	Girls	Boys	Girls
Addis Ababa	1387	1500	162	166	220	267
Afar	756	603	141	114	34	22

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Amhara	9309	9317	1913	1887	1711	1572
Oromia	12,883	12,934	2709	2670	2734	2099
SNNPR	7203	7287	1522	1500	1474	1113
Somali	2268	1950	445	389	157	95
Tigray	2080	2143	443	431	447	436
Total	**35,886**	**35,734**	**7,335**	**7,157**	**6,777**	**5,604**

59. The total male population, in million, in the seven regions is nearly:
A. 13 B. 20 C. 26 D. 35
Hint: Total male population is nearly.
Answer: D

60. The three least populous regions in order are:
A. Afar, Addis Ababa, Somalia
B. Afar, Addis Ababa, SNNPR
C. Addis Ababa, Tigray, Somalia
D. Afar, Ormia, Tigray
Hint: Afar = 756 + 603 = 1359
Addis Ababa = 1,387 + 1500 = 2887
Somali = 2,268 + 1950 = 4218
Answer: A

61. Assuming that the total population of the county at the time the above data is taken is 73 million, what is the share, in percent, of the population of Amhara and Oromia together?
A. 40 B. 50 C. 60 D. 70
Hint: Amhara = 9309 + 9317 = 18,926
Oromia = 12,883 + 12,934 = 25,817
Amhara + Oromia = 18,626 + 25,817 = 44,443
⇒44,443,000 is what percent of 70,000,000
⇒x% of 70,000,000 = 44,443,000

$$\frac{(x)(70,000,000)}{100} = 44,443$$

$$x = \frac{44,443,000}{70,000,000} \times 100\% \approx 60\%$$

Answer: C

62. If the radius of a circle is increased by 30% the area of circle is increased by
A. 30% B. 69% C. 130% D. 169%
Hint: If a quantity is increased by a percentage then the percentage of the quantity is added to the original. Let r be original radius. Thus, R = r + 30% or r = r (1 + 0.3) = r (1.3)

$$\frac{A_2}{A_1} = \frac{\text{Area of circle after increased}}{\text{Area of original circle}} = \frac{\pi R^2}{\pi r^2} = \frac{\pi(1.3r)^2}{\pi r^2} = 1.69$$

$A_2 = A_1(1.69) = A_1 + 0.69A_1 = A_1(1 + 0.69)$
∴ Area of circle increased by 69%.

63. The price of car increased by 2.5%. What is the price of car previously costing Birr 320,000?
A. 800,000 B. 400,000 C. 328,000 D. 360,000
Hint: Percentage increase:

$$\Rightarrow 320,000 + 2.5\% \text{ of } 320,000 = 320,000 + \frac{2.5}{100}(320,000)$$

= 328,000
Answer: C

64. In survey, $\frac{5}{16}$ of people said that they preferred to use self service restaurants. In contrast, $\frac{3}{8}$ of the people indicated that they preferred to be preserved by waiter/ waitress. The remaining 150 respondents stated that they had no clear preference. How many people participated in the survey?
A. 480 B. 390 C. 360 D. 300
Hint: Let x be had no preference, $\frac{5}{16}$ = self service

$\frac{3}{8}$ = by waiter/ waitress

$$\Rightarrow x + \frac{5}{16} + \frac{3}{8} = 1 \Leftrightarrow x = 1 - \frac{11}{6} = \frac{5}{16}$$

Number of people by self service = $\frac{5}{16} = 150$

Number of people by waiter = $\frac{\frac{3}{8}(150)}{\frac{5}{16}} = 180$.

Total number of peoples in survey = 150 + 150 + 180 = 480 people.

65. What amount, when increased by 5% gives 504?

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A. 529.2 B. 336 C. 1008 D. 480

Hint: Let x be amount:

$x + 5\%x = 504 \Rightarrow x(1 + 0.05) = 504$

$\Rightarrow x = \frac{504}{1.05} = 480$

66. In a sale, all prices have reduced by 25%. The sale price of a dress is Birr 350. What was the original price?

A. 262.50 B. 466.70 C. 1400 D. 437.5

Hint: Let x be original price $\Rightarrow x - 25\%$ of $x = 350$

$\Rightarrow x(1 - 0.25) = 350 \Rightarrow x = \frac{350}{0.75} = 466.7$ **Answer: B**

67. In a bowl containing 12 marbles, 6 are yellow and 6 are green. If 2 marbles are picked from the bowl at random, what is the probability that they will both be yellow?

A. $\frac{1}{6}$ B. $\frac{5}{11}$ C. $\frac{1}{2}$ D. $\frac{5}{22}$

Hint: Drawing yellow marble $= \frac{6}{12} = \frac{1}{2}$, but now that marble is no longer in the bowl. So on the second draw the probability of drawing yellow marble is $= \frac{\text{remaining yellow marble}}{\text{remaining marble}} = \frac{5}{11}$.

$\therefore$ The probability of both $= \left(\frac{1}{2}\right)\left(\frac{5}{11}\right) = \frac{5}{22}$. **Answer: D**

68. Quantity comparison:

9 is x percent of 25

y is 25 percent of 80

Column A	Column B
X	Y

A. The quantity in column A is greater.

B. The quantity in column B is greater.

C. The two quantities are equal.

D. The relationship cannot be determined from the information given.

Hint: First translate the statement. $9 = \frac{x}{100}(25) \Rightarrow 9 = \frac{x}{4}$, get $x = 36$

Now translate the 2nd statement, $y = \frac{25}{100}(80) = 20$

$\Rightarrow x > y$ provided that $36 > 20$. **Answer: A**

69. The average of nine number is 16. The average of four of these numbers is 11. What is the average of the other five numbers?

A. 14.46 B. 20 C. 1 D. 5

Hint: Let $x_1, x_2, x_3, \ldots x_9$ are the number.

$\Rightarrow \frac{x_1 + x_2 + x_3 + \ldots x_9}{9} = 16 \Rightarrow x_1 + x_2 + x_3 + \ldots x_9 = 144$

But $\frac{x_1 + x_2 + x_3 + x_4}{4} = 11 \Rightarrow x_1 + x_2 + x_3 + x_4 = 44$

$\Rightarrow (x_1 + x_2 + x_3 + \ldots + x_9) - (x_1 + x_2 + x_3 + x_4) = 144 - 44 = 100$

$\Rightarrow \frac{x_5 + x_6 + x_7 + x_8 + x_9}{5} = \frac{100}{5} = 20$ **Answer: B**

70. The average (arithmetic mean) of a set of six numbers is 32. If a certain number x is removed from the set, the average of the remaining numbers in the set is 25.

A. 13.6 B. 77 C. 67 D. 125

Hint: Let a, b, c, d, e and f are the number.

$\frac{a+b+c+d+e+f}{6} = 32 \Rightarrow a + b + c + d + e + f = 192$

If a certain number x is removed from the set, five numbers left.

$\Rightarrow \frac{a+b+c+d+e}{5} = 25 \Rightarrow a + b + c + d + e = 125$

$= 192 - 125 = 67$ **Answer: C**

71. The price of a jacket was reduced by 20%. During a special sale, the price was discounted another 10%. What was the total percentage discount from the original price of the jacket?

A. 25% B. 30% C. 72% D. 28%

Hint: Let x be the original price of the jacket.

- After 20% discount: $x - x(0.2) = 0.8x$
- Again 10% discount: $0.8x - (0.8x)(0.1) = 0.72x$

So that total percentage discount from the original is given by

$\left(\frac{x - 0.72x}{x}\right) \times 100\% = (1 - 0.72)(100\%) = 28\%$ **Answer: D**

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DIRECTIONS: Question 35 needs to be answered based on the statement, followed by two conclusions labeled I and II. You have to assume the statement is true, then consider the conclusions and decide which of them logically follows beyond a reasonable doubt from the information given in the statement. Then, **blacken** the letter of your choice on the separate answer sheet provided.

35. **Statement**: If all the players play to their full potential, we will win the football match. We have won the football match.
Conclusions: I. All players played to their full potential.
II. Some players did not play to their full potential.

A. Only conclusion I follows
B. Both I and II follows
C. Only conclusion II follow
D. Either I and II follows

Answer: A

1. If $x > y$ compare the two quantities in column A and column B.

Quantity A	Quantity B
30% of x	50% of y

A. The two quantities are equal.
B. Quantity A is less than quantity B.
C. The relationship cannot be determined from the information given.
D. Quantity A is greater than quantity B.

Solution: Since $x > y \Rightarrow 30\%$ of $x = 0.3x$
50% of $y = 0.5y$
i) If $x = -2$, $y = -3$ then A > B
ii) If $x = 3$, $y = 2$ then A < B.
∴ The relationship cannot be determined. **Answer: C**

2.

Quantity A	Quantity B
$(-21)^{12}$	$(-31)^{13}$

A. The two quantities are equal.
B. Quantity A is greater than quantity B.
C. Quantity A is less than quantity B.
D. The relationship cannot be determined from the given information.

Solution: Quantity A: $(-21)^{12} = (-1)^{12}(21)^{12} = 21^{12}$
Quantity B: $(-31)^{13} = (-1)^{13}(31)^{13} = -31^{13}$ from this A > B

Answer: B

3. If s is the number between 0 and 1, which of the following is NOT more than s?

A. $\sqrt{s}$ B. $1/s$ C. $2s$ D. s^2

Solution: $0 < s < 1$ then $s^3 < s < \sqrt{s} < 2s < 1/s$

E.g let $s = 0.5$ which is lies between 0 and 1 then

$s^2 = 0.25, \sqrt{s} = 0.707, 2s = 1$ and $1/s = 2$

$\therefore s^2 = 0.25$ is not more than s **Answer: D**

4. Look at the pattern carefully: 10 20 25 35 40 50 55 then which of the following pairs pattern of numbers comes next?

A. 60, 65 B. 65, 70 C. 60, 75 D. 60, 70

Solution: Pattern: 10 20 25 35 40 55 55 *x y*
by observing the patterns, two even numbers comes together and

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next two odd numbers, the difference between two even numbers and also the difference between two odd numbers is 10. Whereas the difference between even and odd numbers is 5 then. $x = 65$, $y = 70$
$\therefore x = 65$ and $y = 70$ **Answer: B**

5. What is the simplified form of $(1 \div 10^{-1}) \times (-1)^{-10}$?
A. 10 B. 1 C. 0.1 D. 100

Solution: $(1 \div 10^{-1}) \times (-1)^{-10} = \left(1 \div \frac{1}{10}\right) \times \frac{1}{(-1)^{10}} = 10(1) = 10$

Answer: A

6. If $b \neq 0$ and $\frac{a}{5} = \frac{b}{7}$, what is the value of $\frac{a}{b}$?
A. $\frac{7}{5}$ B. $\frac{1}{7}$ C. $\frac{1}{5}$ D. $\frac{5}{7}$

Solution: $\frac{a}{5} = \frac{b}{7} \Rightarrow 7a = 5b \Rightarrow \frac{a}{b} = \frac{5}{7}$

Answer: D

7. Look at the pattern carefully: 32 31 32 29 32 27 32, then which of the following pair of numbers comes next?
A. 29 32 C. 25 32
B. 29 30 D. 25 30

Solution: 32 31 32 29 32 27 32 from the patter
1st, 3rd, 5th, … continue constantly which is 32 whereas 2nd, 4th, 6th
and so on continues like 31, 29, 27, … constantly decreased by 2.
$\therefore 8^{th}$ term is 25 and 9^{th} term is 32 **Answer: C**

8. $1 \div 0.1 \div 0.01 \div 0.001$ is equal to
A. 1,000,000 B. 0.001 C. 100 D. 0.0001

Solution:

$$1 \div 0.1 \div 0.01 \div 0.001 = \frac{\frac{1}{0.1}}{\frac{0.01}{0.001}} = \frac{\frac{1}{0.0001}}{\frac{0.01}{0.001}} = \frac{\frac{1}{10,000}}{\frac{1}{100}} = \frac{1}{10,000} \times 100$$

$$1 \div 0.1 \div 0.001 = \frac{1}{\frac{1}{100}} = 100$$ **Answer: C**

9. What number increased by 3 is equal to 3 less than twice the number?

32. Hassen is standing to the right of Bereket. Beletu is standing on the opposite side of Bereket. Since the opposite of right is wrong, Beletu must be standing on the wrong side of Bereket. Which of the following logical errors has the author of the argument above committed?
A. He has used a single term to mean two different things.
B. He has confused cause the effect.
C. He has provided in factual evidence for his conclusion.
D. He has drawn a general conclusion from an insufficient number of examples.

Answer: A

DIRECTIONS: Read the following assumptions and answer Questions 33 - 34.

Assume J, L, N, M, P, C, A, B, F, E, and R represent persons living in a university dormitory. All live in a building having six floors each having two rooms/apartments. No more than two persons live in any apartment/room. Some rooms may be empty.
L and his roommate live two floors above A and his roommate, C.
J lives alone, three floors below W and two floors below E.
M lives one floor below A and C.
N lives three floors above the floor on which B and F have single rooms.
R and P live in single rooms two floors below M.

Answer: D

33. Which of the following lists the persons named in the correct order, going from the bottom floor to the top?
A. R, F, N, E, L, C C. R, B, M, A, L, N
B. P, B, J, C, N, L D. L, E, A, M, F, R

Answer: D

34. Which of the following is true about where r lives?
A. Second floor, below J or A and C
B. Third floor, above M or E
C. Fourth floor, opposite A and C
D. First floor, below B or F

Answer: D

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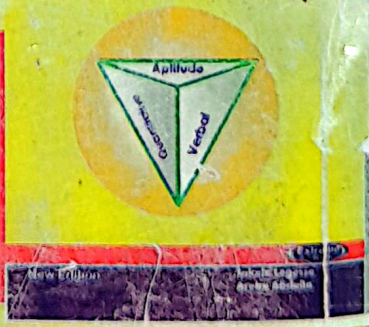

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